

Parameters of counting: Khasi and the crosslinguistic landscape

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Abstract

This paper develops a unified analysis of cardinal expressions, challenging the assumed complementarity between number marking and numeral classifiers (Chierchia 1998; Borer 2005). Drawing on original fieldwork data from Khasi, I show that functional plural marking and individuating numeral classifiers obligatorily co-occur in counting constructions. I argue that this pattern follows once the two are assigned distinct grammatical functions: number marking encodes grammatical number, while classifiers provide the atomic structure required for counting. Building on the Nominal Mapping Parameter (Chierchia 1998), I propose that counting universally requires classifiers and derive cross-linguistic variation from two parameters. The Nominal Mapping Parameter determines noun denotation and the availability of #-based number marking, while a Card/CL Bundling Parameter (Giorgi & Pianesi 1997; Pylkkänen 2002) determines whether Card and CL are projected separately or bundled into a single complex head. Their interaction derives four attested counting systems, exemplified by Mandarin, English, Khasi, and Dëne Sųliné. I further extend the analysis to apparent classifier optionality and mismatches between morphological and semantic plurality, suggesting that these may follow from the interaction of Economy and syntactic-semantic symmetry (Swart & Farkas 2010). The paper argues that classifiers are a universal component of cardinal composition and that number marking and numeral classification reflect independent dimensions of cross-linguistic variation.

Keywords cardinality, classifiers, Khasi, plurality, form-meaning correspondence

1 Introduction

Languages employ two distinct, largely non-overlapping strategies for counting. English marks number morphologically, distinguishing singular from plural forms, as in *one student* versus *two, three, or nine student-s*. Nouns combining with numerals ≥ 2 must therefore bear plural number-marking, rendering (1a) ungrammatical. Mandarin, by contrast, leaves the noun morphologically unchanged but requires a numeral classifier, such as *ge*, between the numeral and the noun (Li 1999), as in (2b). Without a classifier, direct combination of a numeral and a noun is ungrammatical, as in (2a).

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|-----|----|--|----|--|
| (1) | a. | *three student | b. | three student-s |
| (2) | a. | * <i>san xuesheng</i>
three student | b. | <i>san ge xuesheng</i>
three CL student
'three students' |
- (Mandarin, Jiang 2020)

This pattern has been widely taken to reflect a universal generalization: languages that employ numeral classifiers do not have obligatory plural marking (Greenberg et al. 1963; Sanches & Slobin 1973; Aikhenvald 2000). In other words, a language is not expected to require *both* morphological plural marking¹ and numeral classifiers within the same counting system. Several theoretical accounts have offered analyses of grammatical systems that underlie this generalization, arguing that the coexistence of the two systems is either logically (Chierchia 1998) or structurally (Borer 2005) incompatible. Yet, as the range of languages studied has expanded, so too have the exceptions (see §4.5), though none have shown obligatory co-occurrence of the two in their counting constructions.

This paper presents novel fieldwork data from Khasi, an indigenous language of North-East India, which constitutes a clear exception to this generalization. In Khasi, counting plural objects is ungrammatical without both the classifier (CL) *tylli* (3a) and the plural marker *ki* (3b); they *obligatorily co-occur* (3c).

- | | | | | | | |
|-----|----|---|----|--------------------------------------|----|--|
| (3) | a. | * <i>lai tylli ksew</i>
three CL dog | b. | * <i>lai ki-ksew</i>
three PL-dog | c. | <i>lai tylli ki-ksew</i>
three CL PL-dog
'three dogs.' |
|-----|----|---|----|--------------------------------------|----|--|
- (Khasi)

¹ I take the singular noun in number-marking languages to be a number-specified; with the value [+sg]. This very distinction does not hold of Mandarin-type languages, since there is no # projection.

Khasi is particularly noteworthy because both the numeral classifier and the number-marking plural display the characteristic properties associated with these systems, and at the same time their co-occurrence is fully productive, the default strategy, and admits no optionality. This makes Khasi a particularly valuable test case for understanding the roles of numeral classifiers and number marking, and their interaction.

To account for this pattern, and more generally the typology of counting systems, I argue that classifiers are not a language-specific strategy, but a fundamental component of counting across languages, a view independently supported in prior work (Krifka 1995; Wilhelm 2008; Scontras 2013). What varies across languages is not whether classifiers are present, but how they are realized. This variation can be understood in parametric terms; in particular, I propose that two parameters are central. The first concerns the denotation of nouns—the Nominal Mapping Parameter (Chierchia 1998)—which determines how their instantiations are structured and whether they can serve as predicates, arguments, or both. The second is a Bundling Parameter (in the spirit of Giorgi & Pianesi 1997; Pylkkänen 2002), which governs whether the classifier is realized as an independent functional projection or expressed as part of a feature bundle, specifically with a Card(inal) projection (which hosts the numeral).

On this view, the apparent diversity of counting systems reduces to different surface realizations of a shared logic: counting universally requires access to individuated units, and classifiers provide precisely this function. Languages like English obscure this by bundling the classifier function with the numeral, while languages like Khasi realize it overtly as a separate element. Languages further differ in the availability of plural morphology and its interaction with this system. Mandarin, for instance, lacks productive plural marking of the English type, while Khasi combines such plural marking with an overt classifier, yielding the transparent co-occurrence of the two.

This yields a rather neat picture of the diversity of counting systems: we can predict what cardinal expressions² in a language will look like based on the combination of parametric settings that underlie them, and conversely, we can make predictions about the underlying structure of a language’s nominal architecture from how cardinal expressions are realized³. This proposal also

² I have used cardinal expressions and numeral constructions interchangeably. I treat numerals as values of type *n* that serve as arguments for cardinals as well as ordinals (Zabbal 2005, Scontras 2013).

³ See §5 for a discussion on the so-called *optional* classifier languages.

motivates a rethinking of how parametric variation is distributed across grammatical modalities. Rather than treating semantic and morphosyntactic variation as independent, I argue for an integrated approach in which parametric settings across modalities interact within a grammar to give rise to surface variation.

<i>Parameters</i>	NP[+pred, ±arg]	NP[-pred, +arg]
✗ <i>Card/CL</i>	✓#-PL; ✓free CLs Eg. <i>Khasi</i>	✗#-PL; ✓free CLs Eg. <i>Mandarin</i>
✓ <i>Card/CL</i>	✓#-PL; ✗free CLs Eg. <i>English</i>	✗#-PL; ✗free CLs Eg. <i>Dëne Sųliné</i>

Table 1: Parametric typology of cardinal expressions

The remainder of the article is organized as follows. Section 2 reviews the relevant literature on the representation of plurality and the role of classifiers in counting, before turning to the typological generalization concerning their complementarity and its principal syntactic and semantic accounts. Section 3 presents the Khasi pattern and establishes the status of *ki-* as functional number marking and *tylli* as individuating numeral classifiers. Section 4 develops the analysis and derives the resulting typology of counting systems. Section 5 turns to apparent classifier optionality and its implications for the proposed account. Section 6 concludes.

2 Plurals, Classifiers, and their co-occurrence

The complementarity between plural marking and numeral classifiers raises questions about the representation of plurality and the role of classifiers in counting. Before turning to the Khasi pattern, it is therefore important to lay out the broader theoretical landscape in some detail, since the significance of the Khasi facts depends on the precise nature of the two systems and their predicted interaction. I draw on insights from both formal semantics and Distributed Morphology (henceforth DM; Halle et al. 1993; Halle & Marantz 1994) to develop the necessary theoretical background. Of particular relevance are the distinction between functional and lexical plurals, the semantics of plurals and classifiers, and accounts that derive their complementarity.

2.1 Plurals

2.1.1 Functional and lexical plurals: form and function

In number-marking languages, argument NPs are specified for number features, typically singular or plural. Following Ritter (1992), these features are taken to occupy a separate functional

projection, NumP (henceforth #P⁴), and are morphologically realized on the noun and, through agreement, on other elements within the DP and clause. Within the Minimalist Program, # carries interpretable number features and participates in *Agree* relations with other functional heads, including D and T/v. Agreement is triggered when an unvalued instance of a feature F enters into a c-command relation with a valued instance of the same feature, as defined in (4) below. Crucially, number features on the noun are interpretable, while their counterparts on the verb are purely formal agreement features that are deleted prior to interpretation at the interface with semantics.

(4) ***Agree***

- (i) An unvalued feature F (a probe) on a head H scans its c-command domain for another instance of F (a goal) with which to agree.
- (ii) If the goal has a value, its value is assigned as the value of the probe.

(Pesetsky & Torrego 2007: 265)

Plural morphology, however, does not constitute a uniform grammatical category. A distinction particularly relevant here is between functional and lexical plurals (Acquaviva 2008)⁵. Functional plurals form part of the grammatical number system: they apply productively across nouns and participate in number agreement. Lexical plurals, by contrast, exhibit idiosyncrasies in their distribution or interpretation. They are often restricted to and distinguish between noun classes, such as animate vs. inanimate with Bangla *-ra* (Dayal 2012), human vs. non-human with Mandarin *men* (Li 1999), and kind vs. sub-kind with Cuzco Quechua *-kuna* (Dayal 2026), and can contribute meanings beyond plurality, including associative interpretations. Crucially, the two types of plural morphology can coexist within a language: Amharic, for instance, has a functional number-marking system alongside lexical plural morphology, a distinction that Kramer (2016) locates at separate structural positions within DM.

In Kramer's analysis, regular number-marking plurals are associated with the functional number head #, while idiosyncratic or lexical plurals are associated with the nominalizing head *n*. Languages may accordingly exhibit #-based plurality, as in English; *n*-based plurality, as in Halkomelem Salish (Wiltschko 2008); or a split system involving both heads, as in Amharic (illustrated in 5). The existence of double plurals demonstrates that regular and irregular plural morphology are not

⁴ I use # in order to preserve the distinction between numerals and number marking maintained throughout this paper

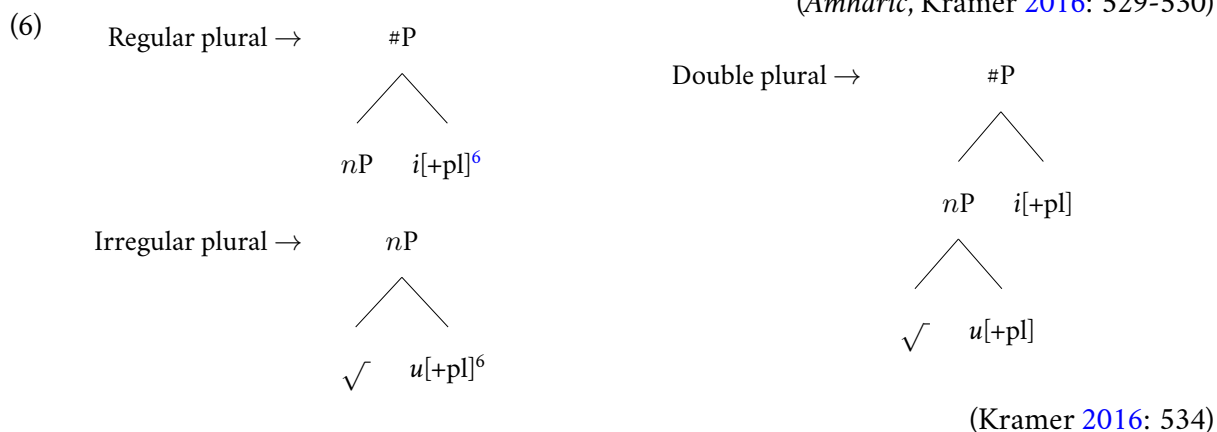
⁵ This distinction broadly corresponds to the distinction between inflectional and non-inflectional plurality in Wiltschko (2008)

merely competing exponents of a single syntactic position, but correspond to *two distinct structural loci* (see also Mathieu & Zareikar 2015 for a three-way distinction of plurals).

(5)

	Singular	Irregular plural	Regular plural	Double plural	gloss
a.	mämhir	mämhir-an	mämhir-ot[tʃ]	mämhir-an-ot[tʃ]	‘teacher’
b.	k'al	k'al-at	k'al-ot[tʃ]	k'al-at-ot[tʃ]	‘word’
c.	mäs'haf	mäs'ahift	mäs'haf-ot[tʃ]	mäs'ahift-ot[tʃ]	‘book’
d.	wäyzäro	wäyzäzirt	wäyzär-ot[tʃ]	wäyzäzirt-ot[tʃ]	‘lady’

(Amharic, Kramer 2016: 529-530)



The significance of this distinction for present purposes is that *n*-based plurality does not entail that a language has number agreement; nor, in a split system like Amharic, does it independently determine DP number (Kramer 2016). Both are tied to #. This division of labor is central to understanding the representation of plurality, the triggering of number agreement, and the interaction between plurals and classifiers. The crucial question here is whether Khasi *ki* is a (functional) #-plural or a (lexical) *n*-plural.

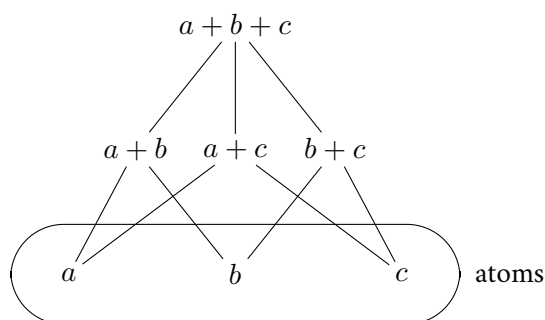
2.1.2 The (un)markedness of singular/plural marking

The most influential semantic account of number, due to Link (1983), formalizes the domain of individuals as a structure of atomic units and the sums they form, organized as a semi-lattice under a part-whole relation, represented in (7) below. A central component of this system is the closure operation, often represented as the *-operator, which maps a set of atomic individuals to its closure under sum formation. Applied to a singular predicate, this operation yields a plural denotation that contains both atomic individuals and all possible sums formed from them. In this way, plurality does not introduce a distinct type of entity, but systematically extends the denotation of

⁶ The notation follows Kramer (2016): *i*[+pl] marks the interpretable plural feature on #, while *u*[+pl] marks the plural feature on *n*, which does not independently determine the number value of the nominal.

a predicate to include structured plural individuals.

(7)



Link's treatment of plurals underlies the *inclusive* view, according to which plural predicates range over both atoms and sums. This is borne out by cases where a plural form can range over a single individual. Thus, *Tony does not have children* is false if Tony has one child, and *Tony has never seen unicorns* if he has seen even one. Yet plural forms are infelicitous with canonically singular referents: *Tony got hit on the nose/#noses* and *Tony has never met his biological mother/#mothers* are infelicitous under the default assumption of a single nose and biological mother. Such facts have motivated the *exclusive* view, according to which plural nouns denote sums but not atoms (Chierchia 1998, though not Chierchia 2010; Chierchia 2021). The contrast between the inclusive and exclusive views has in turn been analyzed as arising from the interaction of semantics and pragmatics. Sauerland (2003) and Sauerland et al. (2005), following Krifka (1989), treat plurals as semantically unmarked and the singular as presupposing atomic reference. The exclusive reading is derived via *Maximize Presupposition!* (Heim 1991), which rules out the plural form in contexts where the presuppositions of the singular are met.

(8) a. $\llbracket \text{SING} \rrbracket(x)$ is defined only if there is exactly 1 x

$\llbracket \text{SING} \rrbracket(x) = x$ whenever it is defined

b. $\llbracket \text{PLUR} \rrbracket(x)$ is always defined

$\llbracket \text{PLUR} \rrbracket(x) = x$ whenever it is defined

(Sauerland et al. 2005: 3)

Hungarian plural *-k* further illustrates the dissociation between morphological and semantic number (Swart & Farkas 2010). Like English *-s*, *-k* marks plural reference and triggers plural agreement (9), both characteristic of functional plurals. However, it is incompatible with determiners/modifiers which independently express plural reference (9). Thus, morphologically singular forms in Hungarian can have sum reference when this is independently enforced by the determiner.

- (9) a. *a gyerekek {elmentek/ *elment}* b. *három/ sok {gyerek/ *gyerekek}*
 the child.PL leave.PAST.PL/ leave.PAST three/ many child/ child.PL
 Int: ‘The children left.’ Int: ‘three children. / ‘many children.’

(Hungarian, Swart & Farkas 2010: 9–10)

Drawing on McCawley (1993) and Horn (1989), Swart & Farkas (2010) advocate for the parsimonious nature of natural language grammar: marked grammatical forms are semantically potent whereas unmarked forms are semantically neutral with respect to the presence of a property (a generalization termed *Horn’s pattern* in Van Rooij 2004). They propose a typological generalization: languages with a morphological singular/plural distinction may allow morphologically singular forms to have sum reference when sum reference is entailed by the determiner. The choice between singular and plural morphology is derived through Optimality-Theoretic competition between economy and correspondence between morphological form and semantic interpretation. This interaction between morphological markedness and semantic plurality will be central to §5.1, where I appeal to this framework to account for languages with so-called ‘optional’ classifiers.

2.1.3 Classifiers: atomizers vs. measure functions

As discussed in §, classifier languages employ an additional functional element, a numeral classifier, in counting constructions. The role of this element has received two distinct analyses: as atomizers or as measure functions. On the atomizer view, the role of classifiers follows from the denotation of bare nouns. While nouns in number-marking languages denote predicates of individuals, bare nouns in classifier languages have been argued to denote *kinds* (Krifka 1995, Chierchia 1998). For example, in Mandarin, common nouns such as *yintao* ‘cherry’ and *gou* ‘dog’ denote the *cherry* and *dog* kinds, respectively, rather than sets of individual cherries or dogs. As Chierchia notes, kinds are isomorphic to sum-closed properties. A sum-closed property, of type $\langle s, et \rangle$, is true of every instance of the property in a world, while a kind, of type $\langle s, e \rangle$ (henceforth e^k), picks out the maximal sum of those instances in each world. Consequently, atomic individuals are not directly accessible to operators that require predicates of individuals, such as cardinals; this is precisely the function of classifiers. They are type-shifting operators that map kind-denoting expressions to atomic predicates (Trinh 2011), as formalized in (10).

(10) **Classifiers as atomizers**

- a. $\llbracket \text{CL}_{\langle e^k, et \rangle} \rrbracket = \lambda k_{e^k} . AT(^{\cup}k)$
- b. $AT(^{\cup}k) = \lambda x_e [^{\cup}k(x) \wedge \forall y_e [(^{\cup}k(y) \wedge y \leq x) \rightarrow y = x]]$ (adapted, Trinh 2011)

An alternative view treats classifiers as measure functions (Krifka 1995; Bale & Coon 2014; Bale et al. 2019; Scontras 2013). Under this approach, numerals denote relations over measures and take a measure function as an argument. A numeral such as *two*, for example, takes a measure function m and a predicate P , and returns those individuals in P whose measure under m is two, as formalized in (12) below. Bale & Coon (2014) provide morphosyntactic support for such an analysis from Ch’ol, where the numeral and classifier form a complex morphological unit, as in (11).

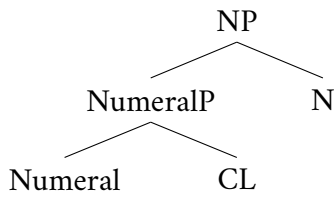
- (11) a. *ux-*(p’ej) tyumuti*
 three-CL egg
 ‘three eggs’ (Ch’ol, Bale & Coon 2014: 5)

(12) **Classifiers as measure functions**

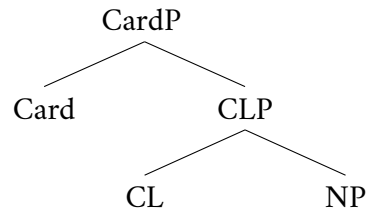
- a. $\llbracket \text{three} \rrbracket = \lambda m \lambda P \lambda x . [P(x) \wedge m(x) = 3]$
- b. $\llbracket \text{CL} \rrbracket = \mu_{\#}$ (Bale & Coon 2014: 7)

Little et al. (2022) argue that both analyses are empirically justified, but characterize distinct types of classifier languages: classifier-for-noun systems, such as Mandarin, and classifier-for-numeral systems, such as Ch’ol. They propose the following structures for the two systems.

(13) a. **CL for numerals**



b. **CL for nouns**



(Little et al. 2022)

Of particular relevance here is that the two classifier systems are not freely interchangeable, but reflect distinct grammatical strategies tied to the semantics of numerals. In a system like Ch’ol, numerals operate over pluralities and require a classifier-supplied measure function; in Mandarin, they operate over sets of atoms made available by prior individuation of the noun. This distinction will be central to the Card/CL Bundling Parameter developed in §4.2.2. We have finally arrived at the issue of plural-classifier co-occurrence, where Khasi presents a particularly distinctive pattern.

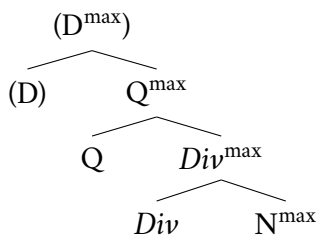
2.2 Co-occurrence

As discussed in §2.1, plurals do not form a homogeneous class, but instead comprise distinct types. Some languages exhibit the co-occurrence of two types of plurals: # and *n*, as in Amharic (Kramer 2016; 5). Similarly, a number of languages with overt classifiers possess effectively lexical plural morphemes, which may even co-occur with classifiers (see Jiang 2020 on Mandarin *men*). Importantly, theoretical accounts of plural-classifier complementarity specifically target #-plurals rather than lexical ones. Two influential approaches derive this restriction: a syntactic account proposed by Borer (2005) and a semantic account developed by Chierchia (1998).

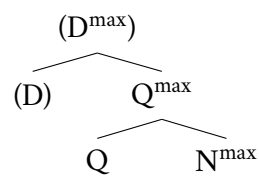
Borer (2005) takes the absence of obligatory plural marking in classifier languages to reflect a deeper structural property of nominal systems, rather than a superficial morphological gap. Specifically, the complementary distribution is not accidental, but follows from the fact that classifiers and plural marking realize the same underlying grammatical function.

Positing an Exoskeletal Model of syntax, Borer claims that all noun denotations are fundamentally mass-like: nouns receive a mass or count interpretation depending on the functional structure in which they occur. The two relevant structures are illustrated in (14) below, where *D* stands for Determiner, *Q* stands for Quantity, *Div* is the ‘divider’ of the noun denotation, and *max* represents the maximal projection for any functional head.

(14) a. Count structure



b. Mass structure



(Borer & Ouwayda 2021: 1)

On this view, nominal roots do not inherently encode atomicity; rather, a functional projection introduces countable units, realized overtly by classifiers in languages like Mandarin and by plural morphology in number-marking languages like English. Classifiers and plural morphology thus instantiate the same dividing function, yielding their complementary distribution: languages realize this function through either classifier or plural morphology, but not both within the same structural domain. This analysis is supported by the Armenian pattern, where both classifiers and plural morphology are available but never co-occur within the same nominal expression.

Within the formal semantics literature, the most prominent account of the complementarity between classifiers and plural marking is due to Chierchia (1998). Unlike Borer, Chierchia derives this contrast from the semantic type of nouns rather than from their syntactic structure. On this view, the complementarity follows from the Nominal Mapping Parameter (see §4.2.1), under which bare nouns in languages such as Mandarin denote kinds rather than predicates. This gives rise to two related consequences. First, since kind-denoting expressions (type e^k) lack accessible atomic structure, they cannot combine directly with modifiers of type $\langle e, t \rangle$ such as cardinals, and therefore require classifiers to shift them into predicates with accessible atomic structure. Second, in predicate-type languages such as English, nouns are borne out of the lexicon as $\langle e, t \rangle$ -type predicates with atomic structure and can therefore combine directly with numerals.

3 Bare nouns, plurals, and classifiers in Khasi

As shown in §1, the system in Khasi is that of co-occurrence, repeated below in (15). This pattern is robust across the Khasic languages: Khasi, Amwi (Varghese 2023), Pnar (Anissa Lamare, p.c.), and Maram, as in (16) below. This indicates a systematicity over an idiosyncrasy of Khasi.

- (16) a. *hynru* ***(tylli)** ***(ki)**-kot
 six CL PL-book
 ‘six books.’ (*Pnar*)
- b. *u* ***(rbe)** ***(i)**-hun-tirme
 two CL PL-child-male
 ‘two sons.’ (*Amwi*)

have been reported to co-occur (see §4.5). However, it is not negligible that in such cases, either the plural or the classifier is non-canonical. As mentioned before, the specific case of concern is the obligatory co-occurrence of (default) functional plurals with (default) individuating classifiers in a language, and the co-occurrence mediating between the numeral and the noun. I argue that Khasi does exactly that: *ki* is a #-plural and *tylli* is a canonical individuating classifier, and that their co-occurrence constitutes a previously undocumented pattern.

3.2 Basic distribution

Khasi nouns are morphologically marked for number and gender in argument positions: singular feminine nouns take *ka-* (17a); singular masculine nouns *u-* (17b); and plural nouns *ki-* (17c), being underspecified for gender.

- | | | |
|-------------------------------|-------------------------------|--------------------------------|
| (17) a. <i>ka-sngi</i> | b. <i>u-tiew-kulap</i> | c. <i>ki-tiew-kulap</i> |
| FEM-sun | MASC-flower-rose | PL-flower-rose |
| ‘sun’ | ‘rose’ | ‘roses’ (Khasi) |

Khasi exhibits subject-verb agreement. In (18a), the singular noun ‘dog’ and the VP ‘barking’ *both* must be marked with the masculine prefix *u-*. Similarly for the plural (18b).

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|--|---------------------------------------|
| (18) a. <i>u-ksew u-dang-wiar</i> | b. <i>ki-ksew ki-dang-wiar</i> |
| MASC-dog MASC-PROG-bark | PL-dog PL-PROG-bark |
| ‘The dog is barking.’ | ‘The dogs are barking.’ (Khasi) |

The unmarked bare noun makes *number-neutral reference* but is restricted to object positions, as with *kot* ‘book’ in (19a). Subjects *must* be number-marked and trigger corresponding agreement on the VP predicate, as with *ka-kynthei ka-thied* ‘FEM-girl FEM-buy’ in (19a). Thus, (19b) is ungrammatical with or without VP agreement; gender or plural marking yields a strictly singular or plural interpretation, respectively.

- | | |
|--|--|
| (19) a. <i>ka-kynthei ka-thied ✓kot</i> | b. <i>*kot (ka)-sngewtynnad</i> |
| FEM-girl FEM-buy book | book FEM-enjoyable |
| ‘The girl bought a book/books.’ | Int: ‘The book is interesting,’ |
| | (Khasi) |

Given Khasi’s English-like number marking, evidenced by subject–verb agreement, its cardinal expressions would be expected to pattern with number-marking languages, as in *three book-s*. This prediction is not borne out (20a). Similarly, classifiers in Khasi have properties typical of Mandarin-like classifier systems, and yet they don’t suffice by themselves (20b).

- (20) a. *ar **ki**-kot
two PL-book
'two books.'
- b. *ar **tylli** kot
two CL book
'two books.'
- (21) a. two book-(*s)
- b. san *(ge) ren
three CL person
'three persons.' (Mandarin)

The data indicate that the noun must bear number marking, as in English, and that a classifier must intervene between the numeral and the noun, as in Mandarin. I present several facts about Khasi plural marking and classifiers that parallel properties commonly associated with number-marking and classifier languages, respectively, to substantiate this assessment.

3.3 *ki-* is a number-marking plural

Number agreement in Khasi holds both within the nominal structure (22) and on the VP (23–24).

- (22) a. **ka-ne** **ka**-kynthei
FEM-DEM.PROX FEM-girl
'This girl.'
- b. **ki-ne** **ki**-kynthei
PL-DEM.PROX PL-girl
'These girls.' (Khasi)

While subject–verb agreement in Khasi parallels agreement systems more generally, its *morphological realization* is analogous to that of Hindi, a number-marking language (23b, 24b). Crucially, such number marking is not characteristic of lexical plurals (or non-inflectional plurals, in the terms of Wiltschko 2008).

- (23) a. **ki**-ksew **ki**-dang-wiar
PL-dog PL-PROG-bark
'The dogs are barking.' (Khasi)
- b. **kutt-e** **bhōk** **rah-e** **hāi**
dog-PL bark PROG-PL AUX.PL
'The dogs are barking.' (Hindi)
- (24) a. **u**-ksew **u**-dang-wiar
MASC-dog MASC-PR-bark
'The dog is barking.' (Khasi)
- b. **kutt-a** **bhōk** **rah-a** **hai**
dog-MASC bark PR-MASC AUX.SG
'The dog is barking.' (Hindi)

Secondly, a robustly attested property of special plurals in classifier languages is their availability in *associative* uses (Nemoto 2005 for Japanese *tati* and Korean *tul*; Jiang 2020 for Mandarin *men*, a.o.). When attached to a proper name, these plurals yield a group denotation 'x and x's associates'. Khasi *ki* does not permit this reading (25); the construction can only refer to a contextually determined set of individuals bearing the same name, i.e., 'the Molly-s of the class'.

(25) **ki-Molly** *ki-thied-kali*

PL-Molly PL-buy-car

Available: ✓ ‘The Molly-s bought cars.’

✗ ‘Molly and her associates bought cars.’ (Khasi)

Thirdly, *ki*-marked NPs are not instances of fake-mass nouns. Mass nouns exhibit different set of properties in Khasi - they do not permit plural marking like in English.

(26) a. ***ki-um**/ ***ki-beer**/ ***ki-phlang**

PL-water/ PL-beer/ PL-grass

Int: ‘Water’/ ‘Beer’/ ‘Grass’

b. **ka-um**/ **u-beer**/ **ka-phlang**

FEM-water/ MASC-beer/ FEM-grass

‘Water’/ ‘Beer’/ ‘Grass’ (Khasi)

Fourthly, *ki*-marked NPs are not instances of *pluralia tantum*, such as English *clothes* or *scissors*, since all plural forms have singular counterparts. More generally, *ki* exhibits none of the properties characteristic of lexical plurals. By Acquaviva’s (2008) criteria, *ki* satisfies all the diagnostics for a functional plural: *obligatoriness*, since it is required for plural reference; *generality*, since it applies across lexical categories, including nouns, adjectives, demonstratives, and verbs; *determinism*, since it is the default means of expressing or ‘marking’ plurality; and lack of *semantic opacity*, since its interpretation is transparently compositional (illustrated in §4.4).

3.4 *tylli* is an individuating classifier

The term *individuating classifier* refers here to classifiers obligatory in counting constructions, such as *ta* in Bangla or *ge* in Mandarin. As is typical of classifier languages, classifier choice varies by noun class. Khasi has two such classifiers: the default *tylli* (27a) and *ngut*, restricted to human nouns such as *girl*, *boy*, *woman*, and *man* (27b).

(27) a. *saw tylli*/ **ngut ki-bilor*

four CL_[−hu]/ CL_[+hu] PL-bottle

Int: ‘four bottles’

b. *saw ngut*/ **tylli ki-kynthei*

four CL_[+hu]/ CL_[−hu] PL-girl

Int: ‘four girls’ (Khasi)

The individuating classifier in Khasi is *not compatible* with mass nouns (28a), as is generally the case in classifier languages (28b). Instead, mass nouns require measure terms or units, such as container expressions like *bilor* ‘bottle’ or conventional units like *litar* ‘liter’, as in (29) below.

(28) a. *hynriew tylli ka-um*

six CL FEM-water

Int: ‘six portions of water’ (Khasi)

b. *chhaw ta rokto*

six CL blood

Int: ‘six portions of blood’ (Bangla)

- (29) a. *saw bilor u-beer*
four bottle MASC-beer
‘four bottles of beer’
b. *ar litar ka-um*
two liter FEM-water
‘two liters of water’ (Khasi)

Another characteristic property of Khasi classifiers is that they do not recur: a noun combines with exactly one classifier in a counting construction (30–31). As Jiang (2020) argues, this non-recursion constraint indicates that classifiers occupy a unique structural position within the nominal domain rather than forming freely iterable measure expressions. This further suggests that *tylli* is not vacuous, but has a well-defined grammatical function.

- (30) a. **san ge ge ren*
three CL CL person
b. **san ge bang yintao*
three CL pound cherry (Mandarin)

- (31) a. **lai tylli ngut ki-kynthei*
three CL CL.HUM FEM-book
b. **lai tylli tylli ki-kot*
three CL CL FEM-book (Khasi)

To sum up, the Khasi data establish that *ki* and *tylli* fulfill distinct grammatical functions, as a plural marker and classifier, respectively, yielding a typologically distinct system. While several languages exhibit some form of plural–classifier co-occurrence, none replicate the Khasi pattern. I return to these languages in §4.5; for now, I turn to the parametric account that derives the Khasi system.

4 A universal CLP account

4.1 Proposal

I propose that counting requires classifiers in all languages, a view also motivated by Krifka (1995), Wilhelm (2006), and Scontras (2022). Crosslinguistic variation in counting can be derived from two parameters. The first, the **Nominal Mapping Parameter** (Chierchia 1998), concerns the denotation of nouns and determines whether a language projects # or not. The second concerns the morphosyntactic realization of cardinals and classifiers: whether the classifier is projected as an independent functional head (CL) or realized as part of a complex Card/CL head listed in the pre-syntactic lexicon. I take this distinction to be regulated by a **Bundling Parameter** (Giorgi & Pianesi 1997; Pylkkänen 2002). For instance, English instantiates a bundled Card/CL head, whereas

Khasi projects CL independently; Mandarin and Khasi, in turn, differ in the existence of number marking.

Following Ionin & Matushansky (2006), I treat Card as a function that takes two inputs: a numeral of type n and a predicate P . It returns those plural individuals x for which there exists a partition S of x into exactly n non-overlapping cells, each satisfying P , as defined in (32) below.

(32) **Definition of Card**

- a. $\llbracket \text{Card} \rrbracket = \lambda P \lambda n \lambda x. \exists S [\Pi(S)(x) \wedge |S| = n \wedge \forall s \in S P(s)]$
 - b. $\Pi(S)(x) = 1$ iff S is a cover of x , and $\forall z, y \in S [z=y \vee \neg \exists a [a \leq z \wedge a \leq y]]$ (Forbidding that cells of the partition overlap ensures that no element is counted twice.)
 - c. A set of individuals C is a cover of a plural individual X iff X is the sum of all members of C : $+C = X$
- (Ionin & Matushansky 2006:318)

As Ionin & Matushansky (2006) argue, the atomicity of the nominal input is crucial. The partition S is the witness for the cardinality of the plural individual x : the condition $|S| = 2$ counts the number of partition cells, not the number of possible sums in the domain. If Card were allowed to apply to a non-atomic or already pluralized predicate, then a larger plural individual such as $a + b + c$ could incorrectly satisfy *two boys*, since it can be partitioned into two non-overlapping cells such as $\{a + b, c\}$. This is the *atomic requirement* of cardinals, and I argue that satisfying this requirement is exactly the job of CL, the numeral classifier.

Following Scontras (2013), I assume that number-marking and classifier languages share a Measure structure, but may draw on different functions within it. Scontras distinguishes Card, which atomizes predicates in number-marking languages, from CL, which atomizes kinds in classifier languages. I unify these functions by treating Scontras's Card as CL: a type-flexible atomizing function that applies to either predicates or kinds and returns an atomic predicate. Thus, CL maps a predicate P to $AT(P)$, and a kind k to $AT(^{\cup}k)$, as formalized in (33).

(33) **Definition of CL**

- a. (i) $\llbracket \text{CL}_{\langle et, et \rangle} \rrbracket = \lambda P. AT(P)$
(ii) $AT(P) = \lambda x. [P(x) \wedge \forall y [(P(y) \wedge y \leq x) \rightarrow (y = x)]]$
- b. (i) $\llbracket \text{CL}_{\langle e^k, et \rangle} \rrbracket = \lambda k. AT(^{\cup}k)$

$$(ii) AT(^{\cup}k) = \lambda x. [^{\cup}k(x) \wedge \forall y [(^{\cup}k(y) \wedge y \leq x) \rightarrow (y = x)]]$$

4.2 Parameters

4.2.1 Nominal Mapping Parameter

Chierchia's (1998) Nominal Mapping Parameter (NMP) distinguishes languages in which nouns are borne out of the lexicon as predicates, as in English, from those in which they are borne out as kinds, as in Mandarin. This distinction captures whether bare nouns function as predicates, arguments, or both. In Mandarin, kind-denoting bare nouns can function directly as arguments; in English, predicative bare nouns can type-shift to kinds, notably as bare plurals, and thereby function as arguments⁷. In Italian, by contrast, bare nouns cannot freely type-shift to kinds and are consequently restricted as arguments, occurring in object position and only under limited conditions in subject position. Chierchia formalizes this typology in terms of two parameters, NP[±pred] and NP[±arg], whose combinations derive the observed differences across languages such as English, Mandarin, and Italian, as in (34) below.

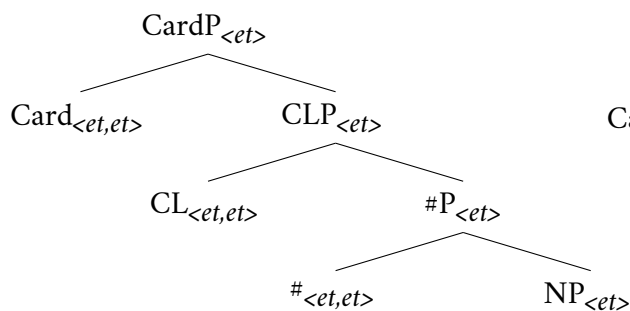
(34) Four logical categories of nominal systems

- | | | | | | |
|-------------------|---|----------|-------------------|---|--------------|
| • NP[-pred, +arg] | — | Mandarin | • NP[+pred, +arg] | — | English |
| • NP[+pred, -arg] | — | Italian | • NP[-pred, -arg] | — | not attested |

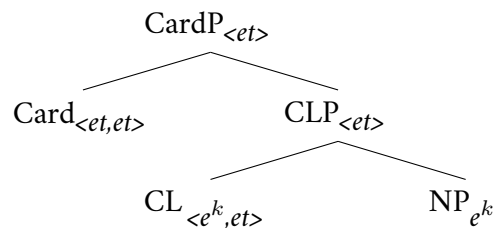
The distinction relevant to the present proposal is between predicate-denoting and kind-denoting bare nouns. I therefore reduce Chierchia's four-way typology to a two-way distinction, represented by the parameter NP[±pred]. I propose that NP[+pred] languages project a #P whose number features are realized through nominal number marking and NP-VP agreement. NP[-pred] languages lack #P and consequently lack both phenomena, although they may retain agreement for other features, such as person in Bangla. The corresponding structures are given in (35). The NMP thus serves as a key determinant of cross-linguistic variation in counting constructions, specifically in predicting whether a language exhibits #-based plural marking (NP[+pred]) or not (NP[-pred]).

⁷ Bare singulars in languages like Hindi and Russian can function as kinds as well as predicates (Dayal 2004). Importantly, English as well as Hindi bare nouns are fundamentally predicative, with singular carrying a [+sg] and plural carrying [+pl].

(35) a. NP[+pred] languages with #P



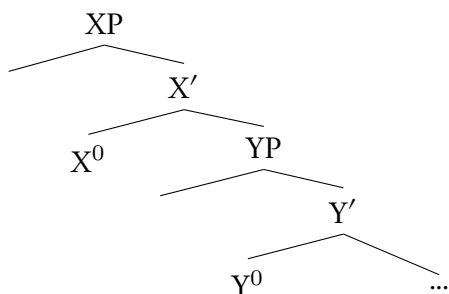
b. NP[-pred] languages without #P



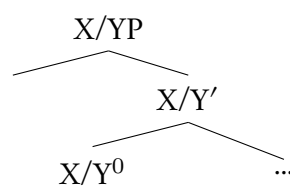
4.2.2 Card/CL Bundling

The distribution of category features across languages has often been formalized in terms of a *bundling* parameter, as in Giorgi & Pianesi's (1997) Feature Scattering Hypothesis. On this view, a set of features [X] and [Y] may be realized either on distinct heads X^0 and Y^0 or on a single bundled head $(X/Y)^8$.

(36) a.



b.



One influential proposal in this domain is Pylkkänen (2002), who introduces a Voice Bundling Parameter under which languages vary in whether Voice is projected separately or bundled with ν (Pylkkänen 2008, Harley 2013, Harley et al. 2017, Coon & Preminger 2012). These heads nonetheless make distinct semantic contributions, with Voice introducing agentive semantics and ν causative semantics (Pylkkänen 2002; Kratzer 1996), even when realized as a single complex head. In Pylkkänen's (2002) account, for example, composition at the bundled V/Cause head proceeds in two successive steps. Extending this approach to the nominal domain, I propose that features encoding individuation (Harley & Ritter 2002) and cardinality may likewise be bundled into a single functional head or distributed across distinct projections (see Šereikaitė 2026 for a cross-categorical bundling account of Voice and n in Lithuanian). What exactly would these features be?

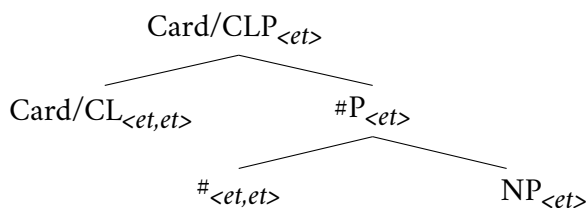
For CL, I argue that the relevant feature is one of individuation, in the spirit of Harley & Rit-

⁸ See Hsu (2021) for detailed exposition and extensions to a syntactic parameter

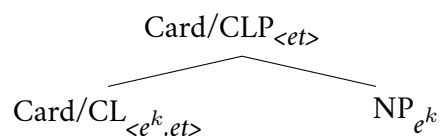
ter (2002), specifically CLASS. For example, the human classifier *ngut* in Khasi (§3.4: 27b) may host the feature [+human]. More generally, classifiers have been argued to categorize nouns according to their inherent properties, such as animacy, shape, and consistency (Croft 1994; Cheng & Sybesma 1999; Aikhenvald 2000). Within a DM framework, one way to capture this is to assume that CL enters into an Agree relation with *n*, acquiring the noun-class features specified there. The CL head then has competing Vocabulary Insertions determined by the features acquired from *n*, although I do not make this explicit in the derivations that follow. As for Card, we could treat it as a feature insofar as it labels a functional category. Alternatively, Card may simply lack discrete feature values altogether.

Structurally, the two features may either be hosted by a single functional head or projected separately. When bundled, Card and CL are realized as a single Card/CL head, yielding systems without overt classifiers; when distributed, they occupy separate projections, yielding a distinct CLP alongside CardP. Cross-linguistic variation in counting constructions thus follows from whether these features are bundled or scattered, rather than from differences in their underlying availability. When projected separately, Card and CL receive distinct phonological exponents, as in Mandarin and Khasi, exemplified in (35) in the previous section. When bundled, a single morphological exponent realizes both functions, as in English and Dëne Sųlíné, shown in (37a) and (37b), respectively.

(37) a. ✓Card/CL language with NP[+pred]



b. ✓Card/CL language with NP[-pred]



In fact, this distinction may explain the two classifier types identified by Little et al. (2022) (see §2.1.3). In some languages, the classifier and numeral are bound morphemes forming a single morphological compound, as in Ch’ol (38). As López (2009) and Little et al. (2022) note, Ch’ol has a vigesimal numeral system, with *-k’al* marking base-twenty numerals and *-p’ej* restricted to numerals below 20.

(38) **Complex numerals in Ch'ol**

1	<i>jum-p'ej</i>	6	<i>wäk-p'ej</i>	11	<i>juñlujum-p'ej</i>	16	<i>wäklujum-p'ej</i>
2	<i>cha'-p'ej</i>	7	<i>wuk-p'ej</i>	12	<i>lajchäm-p'ej</i>	17	<i>wuklujum-p'ej</i>
3	<i>ux-p'ej</i>	8	<i>waxäk-p'ej</i>	13	<i>uxlujum-p'ej</i>	18	<i>waxäklujum-p'ej</i>
4	<i>chäm-p'ej</i>	9	<i>bolom-p'ej</i>	14	<i>chänlujum-p'ej</i>	19	<i>bolomlujum-p'ej</i>
5	<i>jo'-p'ej</i>	10	<i>lujum-p'ej</i>	15	<i>djo'lujum-p'ej</i>	20	<i>juñ-k'al</i>

(Ch'ol, López 2009, Little et al. 2022)

Comparable patterns are attested in Mizo, which has a decimal system in which *sawm* marks base-ten numerals and *pa-* is restricted to numerals below 10 (39). Yoruba, too, shows the same kind of morphological complexity (Ajíbóyè 2010), as illustrated in (39) below.

(39) a. **Complex numerals in Mizo**

1	<i>pa-khat</i>	6	<i>pa-ruk</i>
2	<i>pa-hnih</i>	7	<i>pa-sarih</i>
3	<i>pa-thum</i>	8	<i>pa-riat</i>
4	<i>pa-li</i>	9	<i>pa-kua</i>
5	<i>pa-ngaa</i>	10	<i>sawm</i>

(Mizo, fieldnotes)

b. **Complex numerals in Yoruba**

	Base	m-form	Output	Gloss
a.	èjì	m + èjì	<i>méjì</i>	'two'
b.	èta	m + èta	<i>mẹta</i>	'three'
c.	ẹrin	m- + ẹrin	<i>mẹrin</i>	'four'

(Yoruba, Ajíbóyè & Déchaine 2004, Ajíbóyè 2010)

Krifka (1995) argues that numerals in English encode both a predicativizing function and an atom-accessing function—effectively, a “built-in classifier.” Extending this, Wilhelm (2008) proposes that the contrast between languages such as English or Dëne Sųliné and classifier languages such as Mandarin lies in the semantic composition of numerals, as illustrated in (40).

(40) **Basic vs. human numerals in Dëne Sųliné**

	1	2	3	4	5
Basic	<i>ʔilághe</i>	<i>náke</i>	<i>tághe</i>	<i>dighi</i>	<i>sqlághe</i>
Human	<i>ʔilághì</i>	<i>nádëne</i>	<i>tághì/tanì</i>	<i>dighì/dìnì</i>	<i>sqlághì</i>

(Dëne Sųliné, Cook 2004, Wilhelm 2008)

Adopting a Card/CL bundling analysis of cardinal expressions in languages like Ch'ol, rather than the two-classifier-type analysis of Little et al. (2022), yields two desirable consequences. First, noun-class sensitivity in classifiers, such as animacy, suggests structural accessibility between the classifier and the noun, which follows straightforwardly if CL projects between NP and CardP. Crucially, this sensitivity is not restricted to ‘classifier-for-noun’ languages like Mandarin. As

shown in (41) below, noun class in Ch'ol determines classifier choice: *-p'ej* for inanimate and generic nouns, *kojty* for animals, and *tyikil* for humans, among others (López 2009). A comparable pattern occurs in the Dëne Sųliné numeral system, illustrated in (40) above.

(41) **Noun-sensitivity in Ch'ol numeral classifiers**

Form	Used to count	Examples	Translation
<i>-p'ej</i>	Inanimate/generic	<i>ux-p'ej juñ</i>	'three books'
<i>-kojty</i>	Animals	<i>ux-kojty mis</i>	'three cats'
<i>-tyikil</i>	People	<i>ux-tyikil x'ixik</i>	'three women'
<i>-k'ej</i>	Flat round objects	<i>ux-k'ej waj</i>	'three tortillas'
<i>-ts'ijty</i>	Long things	<i>ux-ts'ijty tye'</i>	'three trees'
<i>-bujch</i>	Seated/propped up things	<i>ux-bujch bux</i>	'three (propped up) bottles'

(Ch'ol, López 2009, Little et al. 2022)

Second, Ionin & Matushansky (2006) highlight empirical challenges for analyses that treat cardinals as occupying a Spec position on the extended NP spine. In languages such as Russian and Inari Sami, lower numerals assign case to their sister nouns (42, 43), suggesting that cardinals cannot be maximal projections in Spec. Under standard assumptions in both GB and Minimalist frameworks that maximal projections do not assign Case, these facts indicate that cardinals must occupy a head position rather than the specifier of another projection.

- (42) a. *četyre šagà*
four step-PAUC
'Four steps'
- b. *šest' šagov*
six step-GEN.PL
'Six steps'
- (Russian, Ionin & Matushansky 2006)

- (43) a. *kyehti/ kulmâ/ nelji/ vittâ/ kuttâ päärni*
two/ three/ four/ five/ six child-ACC.SG
'two/ three/ four/ five/ six children'
- b. *čiččâm/ kâvci/ ovce/ love/ ohtnubáloh/ kyehtnubáloh/ čyeti... pärnid*
seven/ eight/ nine/ ten/ eleven/ twelve/ hundred child-PART.SG
'seven/ eight/ nine/ ten/ eleven/ twelve/ hundred children'
- (Inari Sami, Ionin & Matushansky 2006)

These observations suggest a broader generalization: Card and CL need not be realized as distinct heads, but may instead form a single morphologically and semantically complex unit. In the proposal developed here, this corresponds to the bundling parameter, which predicts that in

such systems numerals (typically hosted in Card) surface as morphologically complex forms. The interpretive effect of the bundled Card/CL head could be modeled in two ways. One approach, following Pylkkänen (2002), derives interpretation compositionally at LF via two successive steps, each corresponding to the semantics of an individual head. Alternatively, the bundled head can be treated as a single composite function that integrates both contributions. In the absence of decisive empirical evidence favoring either approach, I remain agnostic in principle; for expository simplicity, however, I adopt the latter and formalize the semantics of the bundled Card/CL head in (44) below. Card/CL is type-flexible, in the same way as CL.

(44) **Definition of Card/CL**

- a. (i) $\llbracket \text{Card/CL}_{\langle et, \langle n, et \rangle} \rrbracket = \lambda P \lambda n \lambda x. \exists S [\Pi(S)(x) \wedge |S| = n \wedge \forall s \in S [AT(P)(s)]]$
(ii) $AT(P) = \lambda x. [P(x) \wedge \forall y [(P(y) \wedge y \leq x) \rightarrow (y = x)]]$
- b. (i) $\llbracket \text{Card/CL}_{\langle e^k, \langle n, et \rangle} \rrbracket = \lambda k \lambda n \lambda x. \exists S [\Pi(S)(x) \wedge |S| = n \wedge \forall s \in S [AT(^{\cup}k)(s)]]$
(ii) $AT(^{\cup}k) = \lambda x. [^{\cup}k(x) \wedge \forall y [(^{\cup}k(y) \wedge y \leq x) \rightarrow (y = x)]]$

The two parameters, Nominal Mapping and Card/CL Bundling, predict four counting strategies, all of which are attested across languages discussed in the literature, Khasi being one of them: (i) free classifiers with no #-PL marking (Mandarin); (ii) bound classifiers with #-PL marking (English); (iii) free classifiers with #-PL marking (Khasi); and (iv) bound classifiers with no #-PL marking (Dëne Sųliné). These patterns are exemplified in (45) and schematized in (46) below.

- (45) (a) two books
(b) *ar tylli ki-kot*
two CL PL-book
‘two books’ (Khasi)
- (c) *liang ge xuesheng*
two CL student
‘two students’ (Mandarin)
- (d) *náke k’ásba*
two chicken
‘two chickens’ (Dëne Sųliné)

(46) **A new typology of cardinal expressions**

<i>parameters</i>	NP[+pred, ±arg]	NP[-pred, +arg]
$\times$ Card/CLP	$\checkmark$ #-PL; $\checkmark$ free CLs Eg. <i>Khasi</i>	$\times$ #-PL; $\checkmark$ free CLs Eg. <i>Mandarin</i>
$\checkmark$ Card/CLP	$\checkmark$ #-PL; $\times$ free CLs Eg. <i>English</i>	$\times$ #-PL; $\times$ free CLs Eg. <i>Dëne Sųliné</i>

4.3 Semantics and morphosyntax of #PL & #SG

I adopt the view that plural morphemes are semantically contentful. A #PL that occupies # is interpreted as the *-operator following Link 1983. As such I appeal to Horn’s pattern of markedness: *overtly signaled (marked) grammatical forms are semantically potent, in that they encode the presence of a property, whereas unmarked forms are semantically neutral with respect to that property* (Van Rooij 2004).

(47) **Definition of #PL**

a. $\llbracket \#PL \rrbracket^{g,c} = \lambda P. *P$

b. $*P = \{+X : \emptyset \neq X \subseteq P\}$ (Link 1983)

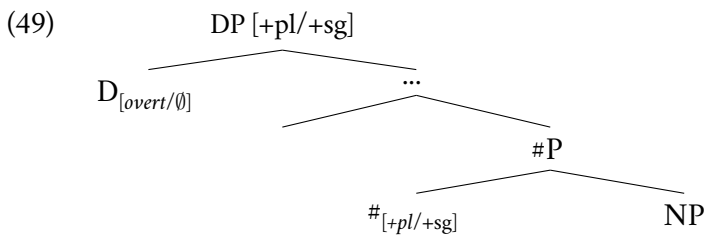
The singular feature, on the other hand, is semantically vacuous: it is an identity function.

However, it carries a presupposition that the predicate already be atomic. Since *AT* has been defined here as an operator from predicates to their atomic sub-predicates, this means that #SG is defined only for predicates *P* such that $P = AT(P)$. In other words, every member of *P* must be an atom relative to *P*.

(48) **Definition of #SG**

$$\llbracket \#SG \rrbracket = \lambda P. P \quad \text{if } P = AT(P), \text{ undefined otherwise.}$$

NPs in NP[+pred] (number marking) languages need to be determined for number - *i*[+pl] or *i*[+sg]. The # head does exactly that. Following Pesetsky & Torrego (2007), I assume that number features are represented in the syntax as part of a broader system of *valuation* and *agreement*, in which features may be either valued or unvalued independently of their interpretability. Within the nominal domain, number is typically valued on the noun while corresponding number features on functional elements (such as D or adjectival heads) are unvalued and receive their value through *Agree*. In the clausal domain, number features on the verb are likewise unvalued and become valued through agreement with the subject DP (49).



It is not trivial that, under this proposal, counting constructions with cardinals ≥ 2 introduce sums at #P only to eliminate them at CLP, seemingly at odds with Economy. I argue, however,

that Economy competes with a principle enforcing a tight correspondence between syntactic and semantic features, defined in (50) following Swart & Farkas (2010).

(50) ***Syntactic and semantic symmetry generalization***

Plural forms should be semantically marked relative to singular forms so as to preserve the correspondence between morphological and semantic markedness seen elsewhere in language.

(Swart & Farkas 2010: p 6:8)

In number-marking languages, NPs must bear number features, $i[+pl]$ or $i[+sg]$, which participate in *Agree* with functional heads such as D and V. The # head thus contributes both syntactically and semantically: it encodes the agreement feature $[+pl]$ and is interpreted as the *-operator, introducing closure under sum formation (Link 1983). The symmetry generalization in (50) aligns these dimensions, requiring semantically plural NPs to bear $i[+pl]$. This interaction determines cardinal composition and the distribution of CL across language types. Where symmetry is enforced, plural marking introduces sums that CL must re-atomize to satisfy Card’s atomicity requirement. The need for CL, however, depends on NP denotation. In $NP[-pred]$ languages, NPs denote kinds and lack atomic structure, requiring CL in all counting constructions. In $NP[+pred]$ languages, CL is required only with plural-marked NPs, whose sums must be re-atomized before combining with Card.

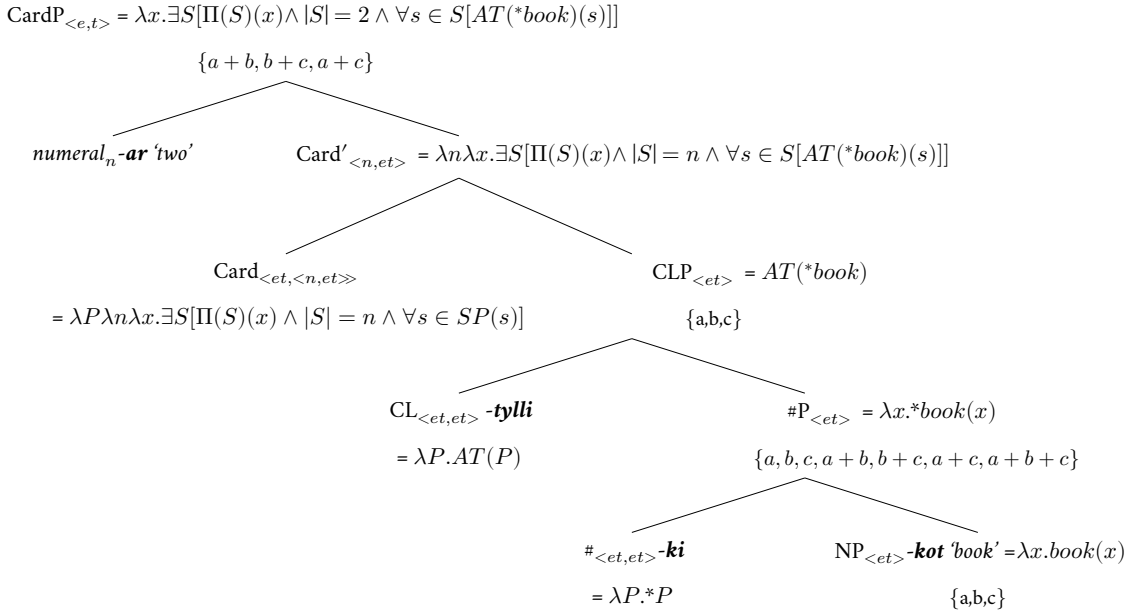
4.4 Derivations

With these definitions in place, let’s look at how counting constructions compose in the four types of languages - (i) Khasi, set to $NP[+pred]$, $\times Card/CLP$; (ii) Mandarin, set to $NP[-pred]$, $\times Card/CLP$; (iii) English, set to $NP[+pred]$, $\checkmark Card/CLP$; and (iv) Dëne Sųliné set to $NP[-pred]$, $\checkmark Card/CLP$. Two clarifications are in order. First, in the trees below, $AT(P)$ abbreviates the atomic subpredicate of P (see §4.1 for definition), as in (51a) below. For kind-denoting inputs, $AT(^{\cup}k)$ denotes the atomic predicate obtained by applying the instantiation operator to the kind and selecting its atomic members, as in (51b). Crucially, $AT(P)$ and $AT(^{\cup}k)$ are not additional conditions imposed by Card on the final plural individual x , but the output of CL: the atomic predicate whose members serve as the cells counted by Card.

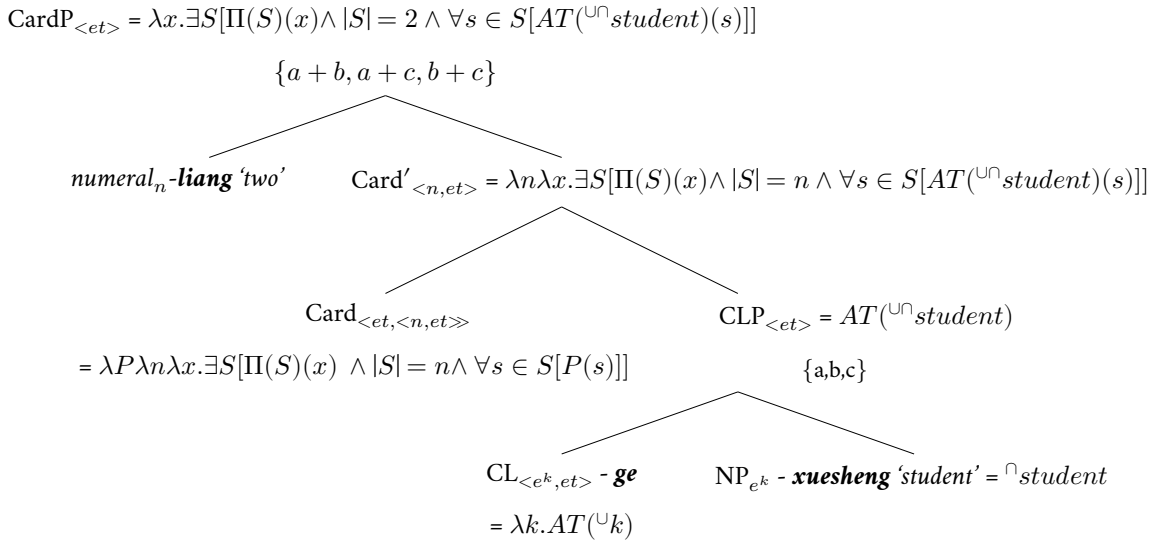
- (51) a. $AT(P) = \lambda x[P(x) \wedge \forall y[(P(y) \wedge y \leq x) \rightarrow y = x]]$
 b. $AT(^{\cup}k) = \lambda x[^{\cup}k(x) \wedge \forall y[^{\cup}k(y) \wedge y \leq x) \rightarrow y = x]]$

Second, at the Card' or Card/CL' level, the expression introduces a partition condition over x , but remains a function from numeral values to predicates until a value for n is supplied. For a given n , Card returns plural individuals x that can be partitioned into n non-overlapping atomic cells satisfying the relevant predicate. Such individuals are therefore n -sums: for $n = 2$, x may be $a + b$, $a + c$, or $b + c$; for $n = 3$, $a + b + c$.

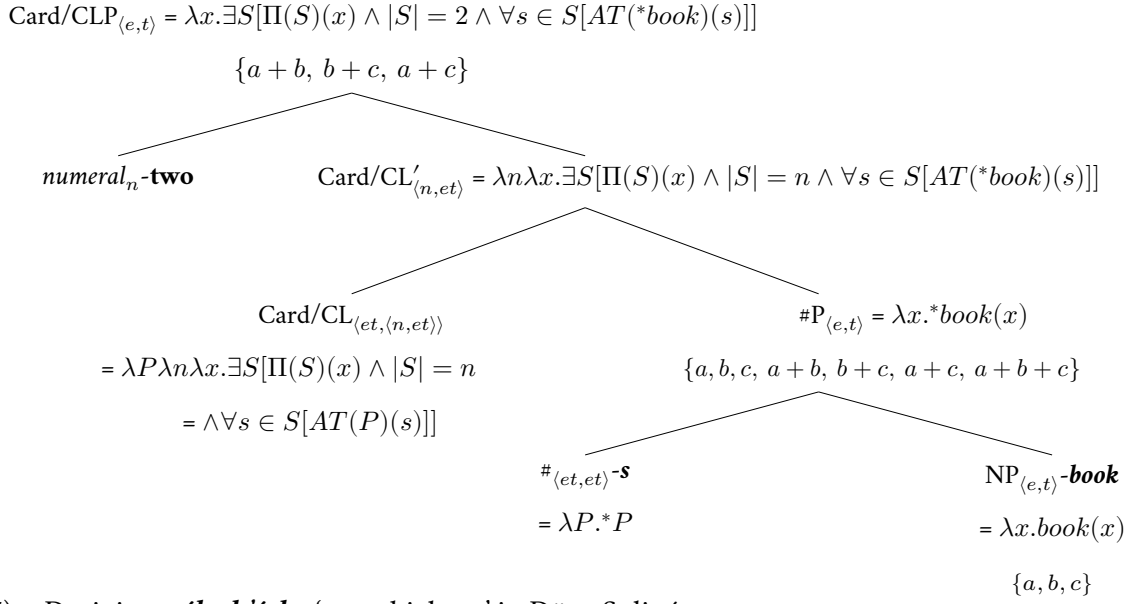
(52) Deriving *ar tylli ki-kot* ‘two books’ in Khasi



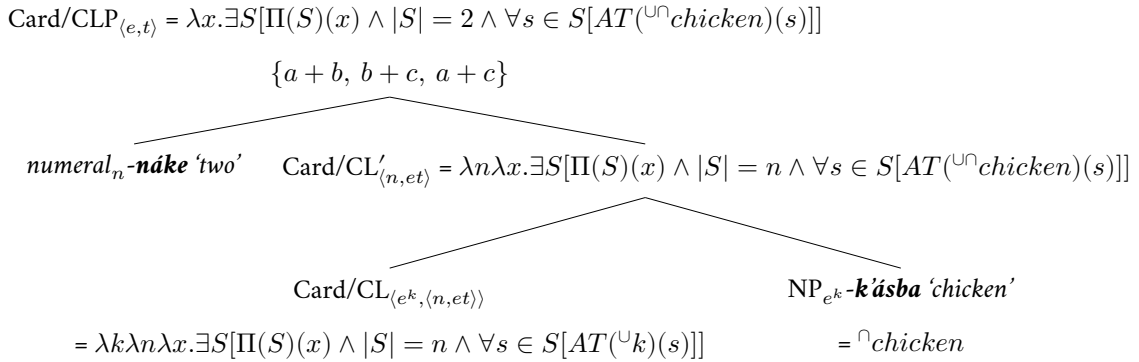
(53) Deriving *liang ge xuesheng* ‘two students’ in Mandarin



(54) Deriving *two books* in English



(55) Deriving *náke k'ásba* 'two chickens' in Dëne Sųliné



4.5 *Some cases of plurals in classifier languages*

In this section, I examine four classifier languages that have been reported to exhibit plural morphology and point out how they differ from the patterns in Khasi *ki*. I adopt four diagnostics for comparison: (i) whether plural marking appears on the main noun, (ii) whether both the plural and the classifier are obligatory in counting constructions, (iii) whether the plural morpheme participates in DP-internal or DP-external (i.e., clausal) agreement, and (iv) whether the plural contributes additional, non-cardinality-related meanings. As demonstrated in (56-59) below, Khasi *ki* satisfies all four criteria of a functional plural.

(56) **PL marked on the noun**

kot / *ka-kot* / *ki-kot*
book / *FEM-book* / *PL-book*
book / *one book* / *books*'

(57) **Number agreement: #-PL**

ki-ne ki-kynthei ki-thied kot
 PL-dem PL-girl PL-buy book
 'These girls bought books'

(58) **CL and #-PL both obligatory**

- | | | |
|---|--|--|
| (a) * <i>lai tylli ksew</i>
three CL dog | (b) * <i>lai ki-ksew</i>
three PL-dog | (c) <i>lai tylli ki-ksew</i>
three CL PL-dog
'three dogs.' |
|---|--|--|

(59) **No special interpretive effect**

- ki-ne ki-khyannah ki-sah ha mawlai*
 PL-DEM PL-student PL-stay in mawlai
 Only available interpretation: 'These students live in Mawlai.' (Khasi)

A commonly discussed type of plural in classifier languages is the so-called associative plural. The defining property of such plurals is that they may attach to proper names or pronouns to denote a group consisting of the named individual and their associates - such as the morpheme *tati* in Japanese (Kurafuji et al. 2004), *ra* in Bangla (Dayal 2014), and *men* in Mandarin (Li 1999). Although these morphemes exhibit language-specific properties, they share the ability to yield associative interpretations. The examples in (60) below illustrate this pattern for Mandarin *men*, which has an additional interpretive effect: of definiteness (60b). Importantly, however, these plurals fail to meet two of the diagnostics satisfied by Khasi *ki*. They are not obligatory in counting constructions; in fact, they are often incompatible in such contexts (60c). Nor do they participate in agreement with the VP, since these are predominantly kind-NP languages that lack systematic number agreement. Instead, their primary contribution is semantic, yielding associative or other readings, such as definiteness in the case of Mandarin *men*.

- | | | |
|---|---|---|
| (60) a. <i>xiaoqiang-men</i>
xiaoqiang-MEN
'XiaoQiang and others' | b. <i>xuesheng-men</i>
student-MEN
'the students' | c. * <i>san ge ren-men</i>
three CL person-MEN
Intended: 'three people' |
|---|---|---|
- (Mandarin, Jiang 2020)

Another type of plural-classifier interaction involves plural morphology that is realized on the classifier itself. I consider three such systems: Vietnamese, Kadiwéu, and Bishnupriya Manipuri. In Vietnamese, the plural morphemes *những* and *các* co-occur with classifier constructions, as shown in (61) below.

- | | |
|---|---|
| (61) a. <i>cuốn sách rất cũ</i> ³
CLF book very old
The book is very old.' | b. <i>những / các cuốn sách rất cũ</i>
PL / PL CLF book very old
The books are very old.' |
|---|---|
- (Vietnamese, Phan & Chierchia 2022:29)

However, a crucial distinction between Vietnamese and Khasi is that plural marking is incompatible with counting constructions, as illustrated in (62) below. Moreover, since Vietnamese is not a number-marking language, these plural morphemes are unlikely to carry interpretable number features. This suggests that they do not function as inflectional number markers comparable to Khasi *ki*, but instead have a different grammatical status.

- (62) ***những**/ ***các** ba cuốn sách
 PL/ PL three CL book
 Intended meaning: ‘The three books’ (Vietnamese, Phan & Chierchia 2022: 29)

A related pattern is found in Kadiwéu, a Waikurúan language. As Sandalo & Michelioudakis (2016) observe, classifiers in Kadiwéu are obligatory in determiner-like environments (including quantifiers, wh-elements, and demonstratives), resembling Mandarin-type systems. Kadiwéu also exhibits multiple plural morphemes, but these differ systematically from Khasi. One plural morpheme (*wa*) attaches to classifiers, while another (*di*) attaches to the noun. The nominal plural *di* contributes meanings beyond simple plurality, such as type-level or varietal readings (63), while regular plurality is often expressed by bare nouns, as is typical of classifier languages.

- (63) a. *apolikaGana-Ga-idi*
 horse-N-PL
 ‘groups of horses / horses of different types’
 b. *ni-nyoGo-di-idi*
 alienable-water-N-PL
 ‘water tanks / water of different types’

(Kadiwéu, Sandalo & Michelioudakis 2016:11)

Kadiwéu has a rich system of plural morphemes, some of which attach specifically to classifiers. Crucially, in contrast to Khasi, plural marking within the DP is not obligatory, and DP-VP agreement is likewise optional. When present, verbal plurality is marked by *o*, which attaches to person agreement morphology, as illustrated in (64).

- (64) *niGi-di-wa ligedema-Ga (o)-nikoteloko libaGa-di*
 DEM.masc-CL-PL frog-n PL-3-sit-on 3-hand-n
 ‘These frogs sit on his hand.’ (Kadiwéu, Sandalo & Michelioudakis 2016: 14)

Bishnupriya Manipuri, an areally proximate language to Khasi, presents yet another pattern. As Akolkar (2023) notes, the plural morpheme *-i* may attach to classifiers, as in *g-i* ‘CL_{go}-PL’ in (65).

Moreover, the example in (65) shows that there is number agreement between the DP and VP; accordingly, the sentence is ungrammatical if the VP is not marked for plural.

- (65) *manu e g-i coktha is-i/(*-e)*
 person DEM.PROX CL_{go}-PL tired be-3PL.PRS/3SG.PRS
 ‘These people are tired.’ (Bishnupriya Manipuri, Akolkar 2023)

However, three systematic differences distinguish this system from Khasi. First, plural marking targets the classifier rather than the noun (66b). Second, pluralized classifiers are not permitted in indefinite counting constructions (67a). Third, the plural morpheme contributes additional semantic content rather than encoding simple cardinality. In (67b) below, the classifier *han* must be marked with the plural *i* if the numeral is marked with the definite morpheme *-o*.

- (66) a. *lerik*
 book
 ‘book(s)’
 b. **lerik-i*
 book-PL
 Intended: ‘books’
- (67) a. *lerik tin han(*-i)*
 book three CL_{han}-PL
 Intended: ‘three books’
 b. *lerik tin-o han*(-i)*
 book three-DEF CL_{han}-PL
 Intended: ‘the three books’
 (Bishnupriya Manipuri, Akolkar 2023)

While these languages exhibit a rich array of plural expressions, they differ from the more familiar number-marking systems found in English-type languages, and their classifiers also serve functions beyond cardinal expressions. Across these cases, plural marking is typically optional in cardinal expressions, does not consistently participate in agreement, and often contributes additional semantic effects beyond simple plurality. By contrast, Khasi *ki* systematically marks plurality on the noun, is obligatory in counting constructions, and participates in agreement relations, thereby patterning with canonical number-marking systems.

Language	PL on noun	#PL Agr	CL & PL obl.	Semantic effects
Khasi	✓	✓	✓	✗
Mandarin	✓	✗	✗	✓
Vietnamese	✗	✗	✗	?
Kadiwéu	✓	<i>optional</i>	✗	✓
Bishnupriya Manipuri	✗	✓	✗	✓

Table 2: Crosslinguistic comparison of plural marking in languages with classifiers

The phenomenon of interest, therefore, is not plural–classifier co-occurrence in general, but specifically the co-existence of an English-type number-marking system and a Mandarin-type numeral classifier system within a single grammar; a configuration that, as the parametric account developed above predicts, is licensed only under a specific combination of the NMP and the Card/CL bundling parameter. In the rest of this paper I offer some perspectives on the (apparent) optionality in classifier languages.

5 Pragmatic constraints and the optionality of classifiers

As discussed in §2.2, a number of languages have been described as exhibiting so-called *optional classifiers*, including Armenian (Borer 2005), Turkish (Göksel & Kerslake 2025), Farsi (Gebhardt 2009), and Hungarian (dekan2012profile). This apparent optionality poses a challenge for the parametric account of the morphological realization of Card and CL proposed in the previous section. That is, if the (non-)realization of classifier morphemes results from a parametrically determined bundling of the two heads in the pre-syntactic lexicon, then their presence or absence should not be freely variable. In this section, I develop an analysis in which the observed variation is derived from the ranking of pragmatic constraints within a language (Swart & Farkas 2010). Specifically, I argue that these (NP[+pred]) languages do not project a CL in counting constructions. Consequently, following Sağ 2025, I propose that the morphemes previously analyzed as classifiers are in fact realizations of the Card function.

5.1 Ordering pragmatic principles

As briefly mentioned in §2.1.2, Hungarian exhibits a mismatch between number morphology and singular/plural reference, repeated in (68) below. Swart & Farkas (2010) account for this by advancing a broader typological claim; the *mismatch generalization*, also repeated in (68c) below.

Several languages fall under this class, including Finnish and Welsh, as illustrated in (69).

- (68) a. *három gyerek/ sok gyerek/ több gyerek*
 three child/ many child/ more child
 ‘three children.’/ ‘many children.’/ ‘more children.’

- b. **három gyerekek/ *sok gyerekek/ *több gyerekek*
 three child.PL/ many child.PL/ more child.PL
 Int: ‘three children.’/ ‘many children.’/ ‘more children.’

(Hungarian, Swart & Farkas 2010: 10)

c. **Mismatch generalization**

There are languages with a morphological singular/plural distinction in nominals, where singular forms may have sum reference in case sum reference is entailed by the determiner.

- (69) a. *Yhdeksän omena-a puto-si maa-han* (Swart & Farkas 2010, p 6:12)
 nine-NOM apple-PART.SG fall-PAST-3SG earth-ILL
 ‘Nine apples fell to earth.’ (Finnish; Nelson & Toivonen 2000)

- b. *y tair cath ddu hynny*
 the.PL three.FEM cat.FEM black.SG that.PL
 ‘those three black cats’ (Welsh; Mittendorf & Sadler 2005)

Swart & Farkas (2010) provide an Optimality Theory analysis that derives this typology from the ranking of general pragmatic constraints available across languages—namely, economy and symmetry between syntactic form and semantic reference—which together yield the observed patterns in Hungarian. Building on their approach, I posit two pragmatic constraints that are universally available; these are defined in (70) below.

(70) **Pragmatic constraints on # and sum reference**

- a. *REDUND: Avoid applying #-PL function (*-operation) if sum reference is available otherwise.
- b. M-SYN/SEM: Match semantic plurality (sum reference) of the NP with syntactic plurality (+pl) of the NP.

These constraints may be ranked differently across languages, and it is precisely this variation that distinguishes Hungarian-type NP[+Pred] languages from NP[+Pred] languages such as English or Khasi. With respect to the Bundling parameter, I argue that Hungarian is a ✕Card/CL language, in which Card is projected independently. In (71-72) below, I illustrate how differ-

ences in constraint ranking give rise to the observed variation between Hungarian and Khasi, both NP[+Pred] and ✕Card/CL languages. The key distinction lies in whether the constraint ranking requires #-PL marking on the NP. In Hungarian, *REDUND outranks M-SYN/SEM, allowing a mismatch between sum reference and syntactic number in contexts where sum reference can be derived independently of the *-operator, as in *három gyerek*, where Card alone yields plural interpretation, (71a). However, such a mismatch is not permitted when sum reference can only be derived via the *-operator, as in *a gyerek-ek* (71b). Since Hungarian permits a mismatch between sum reference and syntactic number, and the *-operator need not apply at #P, there is no requirement to project a CL between Card and NP; although a CL function is in principle available, it is not required in cardinal expressions, all else being equal.

(71) Hungarian: *REDUND ≫ M-SYN/SEM

a.	‘three children’	*REDUND	M-SYN/SEM
	☞ <i>három gyerek</i> three child		*
	<i>három gyerek-ek</i> three child-PL	*!	

b.	‘the children’	*REDUND	M-SYN/SEM
	<i>a gyerek</i> the child		*!
	☞ <i>a gyerek-ek</i> the child-PL		

By contrast, in Khasi, M-SYN/SEM outranks *REDUND, enforcing a strict correspondence between sum reference and syntactic number; as a result, the *-operator must apply at #P, and a CL must be projected in cardinal expressions⁹.

⁹ The skeptical reader may justifiably raise an eyebrow at the idea that pragmatic constraints could result in ungrammaticality. What I want to highlight here, however, is that despite the unconventionality of this treatment, the pattern itself is real. Hungarian systematically distinguishes the grammaticality of number-marking plurals in the presence of cardinal and quantifier expressions, and the OT proposal advanced in Swart & Farkas (2010) offers one way of capturing this systematic pattern. I remain partially agnostic about the correct analysis, but this approach is intuitively compelling at present.

(72) *Khasi*: M-SYN/SEM $\gg$ *REDUND

‘three children’	M-SYN/SEM	*REDUND
a. <i>lai tylli ka-khynnah</i> three CL FEM-child	*!	
b. <i>lai tylli ki-khynnah</i> three CL PL-child		*

To see how this plays out compositionally, consider the derivations in (73-74). In (73) the DP *a gyerek-ek* ‘the child-PL’ requires the presence of the plural morpheme *-ek* to yield sum reference; in its absence, sum reference cannot be derived.

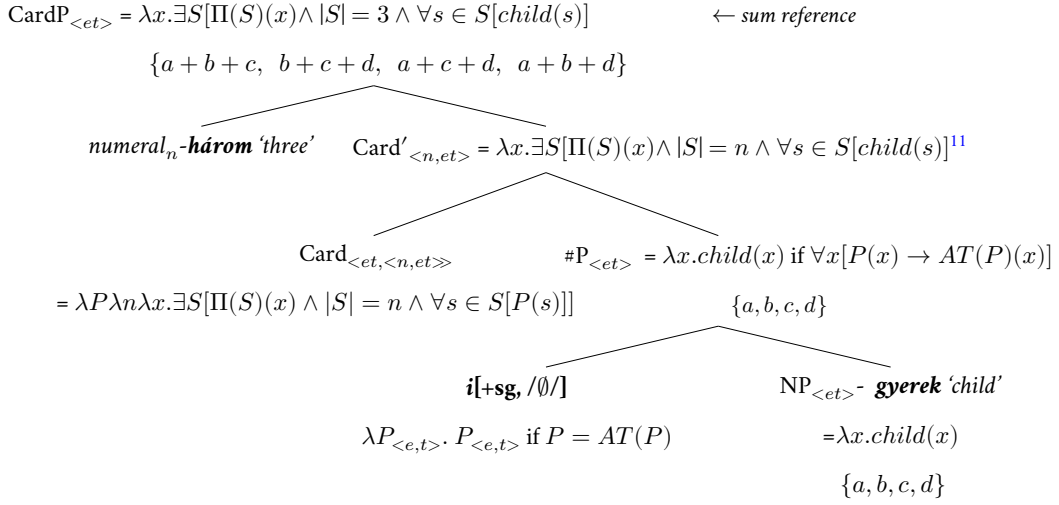
(73) Deriving *a gyerek-ek* ‘the children’ in Hungarian

$$\begin{array}{l}
 \text{DP}_e = \iota x[(\text{Max}(*child))(x)] \text{ if } \exists!x[(\text{Max}(*child))(x)] \\
 \\
 \begin{array}{c}
 \{a + b + c\} \\
 \swarrow \quad \searrow \\
 D^{10}_{\langle et, e \rangle} \text{ -a ‘the’} \quad \#P_{\langle et \rangle} = \lambda x.*child(x) \quad \leftarrow \text{sum reference} \\
 \\
 = \lambda P.\iota x[(\text{Max}(P))(x)] \quad \{a, b, c, a + b, b + c, a + c, a + b + c\} \\
 \text{if } \exists!x[(\text{Max}(P))(x)] \quad \begin{array}{c} \text{i[+pl, /-ek/]} \quad \text{NP}_{\langle et \rangle} \text{ - gyerek ‘child’} = \lambda x.child(x) \\ \lambda P_{\langle et \rangle}.*P_{\langle et \rangle} \quad \{a, b, c\} \end{array}
 \end{array}
 \end{array}$$

By contrast, in (74) *három gyerek* ‘three child’, the plural function need not apply, since Card itself suffices to introduce sum reference: it yields the set of all possible three-member sums drawn from the denotation of *gyerek* ‘child’ in the evaluation context. In such cases, the grammar prefers the vacuous singular form, though the resulting predicate is semantically plural in either case. Importantly, CL need not be projected, since *gyerek* denotes an atomic set and is therefore directly compatible with Card.

¹⁰ The definition of D adopted here follows the standard assumption (Sharvy 1980) for plural definites that the *Max* operator extracts the maximal individuals of the semi-lattice and ι then picks out the unique maximal individual if one exists. See Dayal (2026): Chapter 2 for a comprehensive exposition on this.

(74) Deriving *három gyerek* ‘three children’ in Hungarian



In Khasi and English, by contrast, the constraint enforcing correspondence between plural meaning and plural morphology (M-SYN/SEM) outranks the economy constraint (*REDUND), ensuring that semantically plural predicates are also morphologically marked as plural. I suggest that this ranking extends to Finnish, Welsh, and similarly to Turkish. However, this leaves open the status of the morpheme *tane* in Turkish, often analyzed as an optional classifier, and I turn to this issue next.

5.2 The apparent optionality of CL

5.2.1 Turkish

Turkish presents a case in which a classifier-like element appears “optionally” in cardinal expressions (75). In addition, it has a dedicated plural marker, *-lar*, which, as in Hungarian, is ungrammatical in cardinal expressions, whether or not *tane* is present, as in (75b) below.

- (75) a. *bir (tane) kitap*
 one ?CL book
 ‘one book’
- b. *iki (tane) kitap(*-lar)*
 two ?CL book-PL
 Int: ‘two books’
 (Turkish, Sağ 2025:910)

However, as Sağ (2025) argues, the distribution of *tane* is in fact predictable. Its presence has systematic interpretive consequences: numeral constructions without *tane* permit both definite and indefinite readings, whereas those with *tane* are restricted to indefinite interpretations. Under Sağ’s proposal, Turkish lacks overt definite determiners, and (modified) NP arguments receive

¹¹ See §4.4 for details on how the atomic predicate supplied by CL is converted by Card into a set of *n*-sums at this stage of the derivation.

definite interpretations via covert ι -type-shifting. This type-shift, however, is unavailable in cardinal expressions containing *tane*. As the examples in (76) show, *tane* is only compatible in contexts where an indefinite interpretation of the cardinal expression *üç elma-sın-ı* ‘three apple-3SG.POSS-ACC’ is possible, as in (76b), where it may refer to any three of Sevgi’s four apples. By contrast, *tane* is infelicitous in contexts that force a definite interpretation, as in (76a).

- (76) a. Context: Sevgi has three apples only, familiar to the interlocutors.
Sevgi-nin üç (#tane) elma-sın-ı Merve-ye ver-di-m.
 Sevgi-GEN three CLF apple-3SG.POSS-ACC Merve-DAT give-PST-1SG
 ‘I gave Sevgi’s three apples to Merve.’
- b. Context: Sevgi has four apples, not necessarily familiar to the addressee.
Sevgi-nin üç (tane) elma-sın-ı Merve-ye ver-di-m.
 Sevgi-GEN three CLF apple-3SG.POSS-ACC Merve-DAT give-PST-1SG
 ‘I gave three of Sevgi’s apples to Merve.’ (Turkish, Sağ 2025:941)

Sağ proposes that Card, in the sense of Scontras 2014; Scontras 2022, is overtly realized as *tane* in Turkish, specifically as a variant of Card that yields an argumental, *indefinite* interpretation. Its covert counterpart, by contrast, yields a predicative variant with the same cardinality information, which can undergo ι -type-shifting. What Turkish shows is that the grammar is sensitive to operations of certain argument formations - and the realization of Card determines it.

With this in mind, consider the parametric setting for Card/CL in Turkish. At first blush the system is parallel to what we find in Hungarian in that #-PL *-lar* is not compatible in cardinal expressions. However, there are specific environments in which it can occur, such as in the formation of neighborhood names, as shown in (77) below. Crucially, as Sağ (2025) notes, in such cases the cardinal expression can only receive a definite interpretation, and given the incompatibility of *tane* with definite reference, cardinal expressions containing the plural *-lar* cannot co-occur with *tane*.

- (77) a. *yedi cüce-ler*
 seven dwarf-PL
 ‘the seven dwarfs’
- b. *beş yüz ev-ler*
 five hundred house-PL
 ‘the five hundred houses’ (the name
 of a neighborhood)

(Turkish, Sağ 2025: 949)

That is, the #-PL is compatible in cardinal expressions; thus, a CL function must mediate the for-

mation of cardinal expressions when the NP is marked with #-PL. In light of this data, I contend that Turkish is set to ✓Card/CL. In #-SG-marked cardinal expressions, the CL function applies (vacuously) to the atomic set, specifically as a result of its being bundled with Card in Turkish. As for the constraint ranking, constructions such as (77) in Turkish violate the high-ranked *REDUND, unlike in Hungarian. In light of this, I offer a revision of the constraint definition, illustrated in (78) below.

(78) **Pragmatic constraints on # and sum reference** (revised)

- a. *REDUND: Avoid applying #-PL function (*-operation) if sum reference is available otherwise, unless #-PL has significant interpretive effects beyond *-operation.
- b. M-SYN/SEM: Match semantic plurality (sum reference) of the NP with syntactic plurality (+pl) of the NP.

With respect to the Card/CL setting for Hungarian, the evidence is not definitive, since *-ek* does not encode definiteness effects in the way that Turkish *-lar* does. For now, I maintain my previous claim that Hungarian is set to ✗Card/CL, pending further evidence.

I follow Sağ in analyzing *tane* as the realization of Card_f , while maintaining the formalism developed here. I encode its inability to form definite arguments in the feature specification of Card/CL. Specifically, appealing to allomorphy and contextual allosemy (Marantz 2013), I analyze *tane* as the realization of Card/CL with the synsem feature [-def]. The resulting cardinal expression therefore cannot undergo ι type-shift and be interpreted as a definite argument. The elsewhere realization of Card/CL is $\emptyset$, yielding a case of allomorphy. For the plural, I posit two #-PL morphemes, one specified [+def] and one elsewhere, both realized as *-lar*, yielding contextual allosemy. The V.I. rules and semantics of Card/CL and # are given in (79) and (80) below.

(79) **V.I. rules and semantics of Card/CL in Turkish**

- a. $\text{Card/CL}[-\text{def}] \leftrightarrow \text{tane}$
- b. $\text{Card/CL} \leftrightarrow \emptyset$
- c. $\llbracket \text{Card/CL}[-\text{def}]_{\langle \text{et}, \langle n, e \rangle \rangle} \rrbracket = \lambda P \lambda n. f(\lambda x. \exists S [\Pi(S)(x) \wedge |S| = n \wedge \forall s \in S [AT(P)(s)]])$ (adapted, Sağ 2025: 26)
- d. $\llbracket \text{Card/CL}_{\langle \text{et}, \langle n, \text{et} \rangle \rangle} \rrbracket = \lambda P \lambda n. \lambda x. \exists S [\Pi(S)(x) \wedge |S| = n \wedge \forall s \in S [AT(P)(s)]]$

(80) **V.I. rules and semantics of # in Turkish**

- a. $\#i[+pl, +def] \leftrightarrow lar$
- b. $\#i[+pl] \leftrightarrow lar$
- c. $\llbracket \#i[+pl, +def]_{\langle et, et \rangle} \rrbracket = \lambda P. *P$ if $\exists!x[P(x)]$, undefined otherwise
- d. $\llbracket \#[+pl]_{\langle et, et \rangle} \rrbracket = \lambda P. *P$

Expressions like (77), repeated below as (81), are expected to be incompatible with *tane*, given that there is a feature mismatch within the NP between $\#[+pl, +def]$ and $Card/CL[-def]$.

- (81) a. *yedi (*tane) cüce-ler*
 seven CARD/CL dwarf-PL
 ‘the seven dwarfs’
- b. *beş (*tane) yüz ev-ler*
 five CARD/CL hundred house-PL
 ‘the five hundred houses’ (the name
 of a neighborhood)

(Turkish, adapted, Sağ 2025: 949)

Finally, I address the morpheme *tā* in Farsi¹², whose plural morpheme *ha* has interpretive effects directly responsible for the interpretation of the cardinal expression as an argument. In line with Sağ 2025, I argue that, similarly, Farsi does not have an ‘optional’ classifier. I offer a possible analysis for the distribution of the alleged classifier morpheme *tā* in Farsi in line with my treatment of Turkish *-tane* as given in this section.

5.2.2 Farsi

Farsi is similar to Turkish and Hungarian in that its NPs are marked with #-SG in cardinal expressions, resulting in a mismatch between syntactic number and semantic reference, as illustrated in (82) below.

- (82) a. *se-ta ketab ru-ye miz bud.*
 three-CL book on-EZ table be.PAST.1SG
 ‘Three books were on the table.’

¹² Western Armenian has also been noted as an optional classifier language in the literature (Borer 2005), which is attributed to the morpheme *had*. Following Sağ, I take *had* to not instantiate an independent CL head, paralleling Turkish *tane* and Farsi *ta*. The distribution of *had*, however, is less straightforward. The presence or absence of *had* does not appear to affect the argumenthood of the cardinal expression. Unlike Turkish and Farsi, definite arguments in Western Armenian necessarily require the presence of the overt definite marker that blocks ι -type shifting of the cardinal expression, and such expressions can optionally have the morpheme *had* (eg. 76, Sağ 2025). Similarly with indefinite cardinal expressions - *had* is optional (eg. 75, Sağ 2025). While the precise role of *had* remains unclear, its absence should not be taken to reflect genuine optionality of classifiers. If Western Armenian is likewise set to ✓Card/CL, then both the covert and overt realization of the Card/CL morpheme must be conditioned by other grammatical factors that are not yet fully understood.

- b. *se-ta ketab xærid-æm.*
 three-CL book buy.PAST.1SG
 ‘I bought three books.’

(Farsi, Ghomeshi 2003:59)

Beyond constraint ranking, Farsi and Turkish share several important properties. First, both languages lack overt D morphemes, and definite interpretations arise via ι -type shifting. Second, both exhibit definiteness-dependent Card/CL features, realized either covertly or as an overt morpheme: *tā* in Farsi, as argued in Sağ (2025). She further shows that the distribution of Farsi *tā* is sensitive to whether the cardinal expression is interpreted as definite or indefinite. Specifically, both *tā* and $\emptyset$ are compatible with indefinite interpretations of cardinal expressions, as illustrated in (83), giving rise to the apparent optionality of *tā*.

- (83) *age tu kelās-e man be do (tā) dāneshju komāk kon-i, be nomra-t ezāfe*
 if in class-EZ I to two TA student help-2SG to grade-2SG.POSS increase
mi-kon-am.
 IPFV-do-1SG
 ‘If you help two students in my class, I will raise your grade.’

(Farsi, Sağ 2025:948)

By contrast, only *tā* is compatible with a definite interpretation of the cardinal expression, rendering the null variant $\emptyset$ infelicitous in such contexts; *tā* is therefore not truly optional. This generalization, however, comes with an important caveat: the non-optionality of *tā* is limited to cardinal expressions in which the NP bears the plural morpheme *hā*, as in (85) below. The examples in (84)–(85) form a single discourse context and are naturally uttered in sequence. In (84), the cardinal expression *se (tā) moallem-e zan-(*hā)* ‘three TA teacher-EZ female-PL : three female teachers’ introduces a novel discourse referent and is therefore interpreted as indefinite, rendering *hā* infelicitous.

- (84) a. *Polis dar hāl-e barrasi-e marg-e se (tā) moallem-e zan-(*hā) ast.*
 police in state-EZ investigation-EZ death-EZ three TA teacher-EZ female-PL is
 ‘The police are investigating the death of three female teachers.’

(Farsi, Sağ 2025:950)

In (85), the cardinal expression refers back to the three female teachers introduced in (84), which yields a definite interpretation. In this case, the NP must bear the plural morpheme *hā*, and the morpheme *tā* can *no longer* be omitted.

- (85) a. *Rasāne-hā-ye mahali migooyand se *(tā) zan-hā ke dar dahaye panjah-e*
 channel-PL-EZ local say three TA woman-PL that in 50s-EZ

zendegi-e khod budand.

life-EZ self were

‘Local channels report that the three women were in their 50s.’

(Farsi, Sağ 2025:950)

Crucially, this restriction does not apply when the cardinal expression includes a demonstrative.

In such cases, the expression receives a definite-like interpretation even without plural marking

on the NP, as in (86) below, and *tā* becomes optional again.

- (86) *Rasāne-hā-ye mahali migooyand #(in) se (tā) zan ke dar dahaye panjah-e*
 channel-PL-EZ local say this three TA woman that in 50s-EZ

zendegi-e khod budand.

life-EZ self were

‘Local channels report that these three women were in their 50s.’

(Farsi, Sağ 2025:950)

Building on the framework developed in this paper, I analyze Farsi as a ✓Card/CL language.

On this view, the $\emptyset$ morpheme realizes a Card/CL head specified for a [+indef] feature, while *tā* is inserted as the elsewhere form. The Vocabulary Insertion rules for Card/CL in Farsi are given in (87) below.

(87) **V.I. rules for Card/CL in Farsi**

a. Card/CL[+indef] $\leftrightarrow \emptyset$

b. Card/CL $\leftrightarrow tā$

Semantically, this specified Card/CL head contributes a weak presupposition, akin to the *anti-uniqueness presupposition* associated with demonstratives (Robinson 2005; Dayal & Jiang 2022).

(88) **Semantics of Card/CL in Farsi**

a. $\llbracket \text{Card/CL } [+indef]_{\langle et, \langle n, et \rangle \rangle} \rrbracket = \lambda P \lambda n \lambda s \lambda x. \exists S [\Pi(S)(x)(s) \wedge |S| = n$

$\wedge \forall z \in S[AT(P)(z^{13})(s)]]$ if $\exists s' : s < s'. |P(x)(s')| > 1,$

where *s* is the evaluation situation.

b. $\llbracket \text{Card/CL}_{\langle et, \langle n, et \rangle \rangle} \rrbracket = \lambda P \lambda n \lambda x. \exists S [\Pi(S)(x) \wedge |S| = n \wedge \forall z \in S[AT(P)(z)]]$

Moreover, the plural morpheme *hā* is valued with $i[+pl, -indef]$. The interpretive effects of

¹³ I use *z* here for members of the partition *S*, in order to distinguish them from *s*, the situation variable.

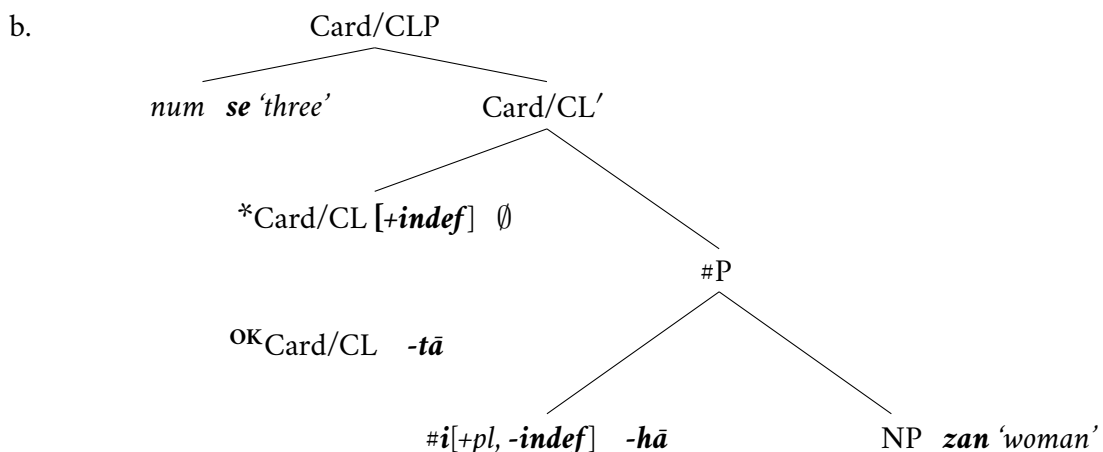
Card/CL are only visible in the presence of *hā*, since it is the covert morpheme that carries the relevant presupposition. I propose that *hā* is the only #-PL available in the grammar, and that it is always specified as *i*[-*indef*]. Semantically, it contributes a presupposition of uniqueness, as shown in (89) below. By contrast, the #-SG function is vacuous, as assumed elsewhere in this paper.

(89) **V.I. rules and semantics of # in Farsi**

- a. $\#i[+pl, -indef] \leftrightarrow hā$
- b. $\#i[+sg] \leftrightarrow \emptyset$
- c. $\llbracket \#i[+pl, -indef]_{\langle et, et \rangle} \rrbracket = \lambda P. *P$ if $\exists!x[P(x)]$, undefined otherwise

This correctly predicts the distribution of *tā* and *hā* in Farsi cardinal expressions. First, the uninterpretable feature [+*indef*] on Card/CL does not itself trigger agreement, but must be matched against the value of [*indef*] at #P. The plural morpheme *hā* realizes $\#i[+pl, -indef]$. When # bears this value and Card/CL is specified as [+*indef*] (realized as $\emptyset$), a feature mismatch arises between # and Card/CL with respect to [*indef*]. This mismatch rules out the null realization and forces insertion of *tā*. This accounts for the obligatoriness of *tā* when the NP is marked with plural *hā*, as illustrated in (90) below.

- (90) a. *se* **(tā) zan-hā*
 three TA woman-PL
 Int: ‘the three women.’

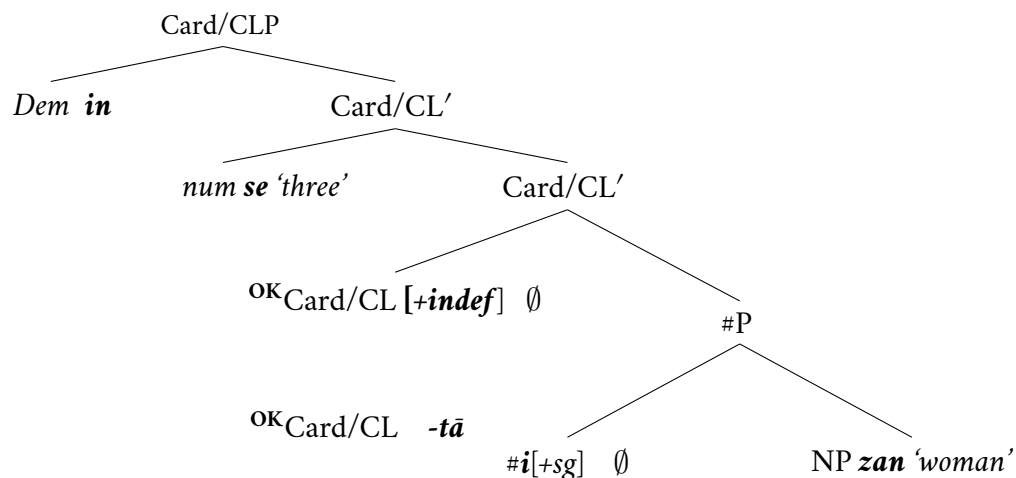


Second, Farsi does not have an overt D head (Sağ 2025), and thus the [+*indef*] feature in Card/CL ($\emptyset$) is not incompatible in demonstrative constructions like (91). Crucially, however, this is only the case when the NP is not marked with the plural *hā*. I argue that the morpheme *in* in Farsi is an NP-modifier, which carries an anti-uniqueness presupposition and a deictic value. In

the ‘definite-like’ use, it gets ι type-shifted. To reiterate, in constructions like (91), there is no feature mismatch if the plural *hā* carrying a $[-indef]$ feature is *not* projected. At the same time, there is no interpretive clash at the LF given that $\text{Card/CL}[+indef]$ simply carries an anti-uniqueness presupposition.

- (91) a. **#(in)** *se* (**tā**) *zan*
 this three TA woman
 Int: ‘these three women’

b.



Stepping back, the apparent optionality in Farsi is only superficial: *tā* realizes the bundled Card/CL head, and its distribution follows from independently determined feature specifications rather than true optionality. We thus get the following picture of crosslinguistic variation in cardinal expressions.

<i>parameters</i>	$\text{NP}[+\text{pred}, \pm\text{arg}]$	$\text{NP}[-\text{pred}, +\text{arg}]$
$\times \text{Card/CL}$	✓ $\#$ -PL; ✓free CLs (<i>modulo ranking</i>) Eg. <i>Khasi, Hungarian</i>	$\times \#$ -PL; ✓free CLs Eg. <i>Mandarin</i>
✓ Card/CL	✓ $\#$ -PL; $\times$ free CLs Eg. <i>English, Mizo, Farsi, Turkish</i>	$\times \#$ -PL; $\times$ free CLs Eg. <i>Dëne Sūliné</i>

Table 3: Updated typology of cardinal expressions

The analysis developed in this section should be understood as one way of conceptualizing these patterns rather than as a complete account of their source. Its central purpose is to show that apparent optionality in plural and classifier marking need not reflect optionality in the underlying semantic operations: it may instead arise from the interaction of morphosyntactic realization

with independently motivated pragmatic constraints. In this sense, the proposal provides a first approximation of how these phenomena might be integrated into the broader typology developed here. A more fully articulated account must explain precisely how the relevant constraints are ranked and, in particular, how a principle such as Economy can give rise to categorical ungrammaticality. I leave this question to future work.

6 Concluding remarks

This paper has developed a unified account of the composition of cardinal expressions and the morphosyntactic variation they exhibit across languages. I have argued that this variation can be derived from two parameters: the Nominal Mapping Parameter, which determines the denotation of nouns and the availability of #-based number marking, and the Card/CL Bundling Parameter, which determines whether the classifier function is independently projected or bundled with Card. Together, these parameters derive a constrained typology of counting systems while maintaining a shared underlying compositional system.

The starting point for this proposal was the apparent complementarity between functional plural marking and numeral classifiers. Novel data from Khasi challenge this generalization: Khasi obligatorily combines a functional #-plural with an individuating numeral classifier in the same cardinal expression. I argued that this pattern follows once the two are assigned distinct grammatical functions. While # introduces grammatical number, CL supplies the atomic predicate required by Card. Their co-occurrence is therefore not only possible but predicted when an NP[+pred] language with #-based number marking independently projects CL. Conversely, languages such as Mandarin lack # but project CL independently, while languages such as English bundle CL with Card. The Khasi pattern thus reveals that number marking and numeral classification are independent dimensions of variation rather than mutually exclusive counting strategies.

Beyond this core typology, the proposal also offers a perspective on variation in the morphological realization of number and classifiers. Building on the interaction between Economy and syntactic-semantic symmetry, I suggested that apparent mismatches between morphological and semantic plurality, as well as so-called classifier optionality, may be derived pragmatically rather than by positing differences in the underlying operations involved in counting. On this view, the classifier function remains semantically necessary, while its overt realization, like that of plural

marking, is subject to independently motivated constraints. The discussion in §5 is intended as an extension of the central proposal rather than a fully developed account; establishing precisely how these constraints interact, particularly how Economy can yield categorical ungrammaticality, remains a question for future work.

The broader consequence is a constrained view of crosslinguistic variation in cardinal expressions. Languages do not differ in whether counting requires individuation, but in how individuation, cardinality, and number are distributed across semantic and morphosyntactic structure. Khasi is particularly revealing because it instantiates the combination excluded by the traditional complementarity generalization: obligatory functional plural marking together with an independently realized numeral classifier. Rather than constituting an isolated exception, the Khasi pattern exposes the independence of the grammatical components involved in counting and shows how their different structural and morphological realizations give rise to the broader typology of counting systems.

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